A GIS based Automated Mapping of Greenville's Heritage Trees and Potential Applications

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Abstract

Trees play a vital role in carbon storage on a global scale, but they also contribute important benefits at the local scale. In urban areas, trees provide ecosystem services that benefit cities, including urban heat island mitigation, air and water pollution reduction, and runoff and flash flooding reduction. The City of Greenville in South Carolina recently passed an ordinance to protect trees on properties with newly permitted developments, including a special emphasis on protecting and encouraging larger, “heritage” trees, despite not having an existing database of protected trees throughout the city. The focus of this research is to use an automated tree detection algorithm to map trees along the streets of the entire city to help protect and monitor the trees and to more equitably enforce the ordinance. LiDAR data and GIS modeling approaches were used to detect individual tree locations and heights throughout the entire city to build an initial database. Multispectral imagery and building geometry data were also used in the process to minimize objects in unvegetated areas from being misclassified as trees. The results show that the automated tree detection algorithm worked well where trees were relatively smaller and separated at the canopy level, but deciduous trees with larger canopy structures were often classified as multiple trees. Field verification is necessary to correct some of these discrepancies. Results from this work can be used to understand the social and environmental inequalities related to green spaces and canopy cover and can drive more data-driven urban trees management strategies.

1. Introduction

As cities continue to grow in population and size, they face pressure between expansion and infill development and preserving their green spaces and urban forests. Cities demand continuous development as they grow, but residents see the value of connections to nature in their urban environment. In addition to challenges brought by increases in population, impacts due to climate change introduce environmental stressors, such as longer and more extreme heat and prolonged precipitation events (Fowler et al. 2021; Prein et al. 2017; Stone, Hess, and Frumkin 2010; Luber and McGeehin 2008; Christensen and Christensen 2004). These compound the environmental challenges cities already face: wastewater treatment, excessive stormwater runoff, air quality concerns, excessive heat in summer absorbed by paved surfaces, etc. (Keeler et al. 2019; Livesley, McPherson, and Calfapietra 2016; Janke, Finlay, and Hobbie 2017; Hobbie et al. 2017; Rötzer et al. 2021).

Cities increasingly realize that they need to implement solutions to these issues, but they also need to be cost effective. Engineered, or constructed, solutions may often be cheaper in the short term, but cities also want resilient solutions that also bolster green spaces, which often benefit their local economy (Keeler et al. 2019). These green spaces include areas with trees, which happen to provide ecosystem system services that support solutions to many of the environmental challenges in the urban context (Keeler et al. 2019; Livesley, McPherson, and Calfapietra 2016).

Urban trees reduce stormwater runoff and can help reduce the magnitude of flash flooding, making the introduction of urban trees a reprieve for overwhelmed water discharge systems that would otherwise need expensive expansions (Rötzer et al. 2021). Trees vary in the amount of precipitation they can intercept, with many trees taking up to 30 minutes before becoming saturated. Through interception and absorption via roots, urban trees retain an average of 18.3% of stormwater, but can sometimes retain as much as 43.7% (Rötzer et al. 2021; Livesley, McPherson, and Calfapietra 2016). Trees planted along bioswales can capture 46-72% of the runoff, but they must be species that can handle fluctuating moisture conditions (Livesley et al., 2016). Urban trees also play a vital role in reducing runoff through rain channeling, where they channel water down their trunks and into the soil, preventing water from landing and flowing on impervious surfaces such as roads and sidewalks. Trees can channel up to 22.8% of precipitation to their trunk during early periods of rainfall (Rötzer et al. 2021). Areas in cities with large amounts of impervious surfaces account for a highly disproportionate amount of stormwater runoff, and replacing some of those impervious surfaces with pervious alternatives that can support the addition of tree cover may help mitigate excess runoff (Dwyer and Miller 1999) by redirecting it trees and groundwater.

Additionally, urban trees reduce water pollution by absorbing nitrate, phosphate, sulfate, carbon, and heavy metals from the water (Livesley, McPherson, and Calfapietra 2016). Turf grass in urban settings is commonly fertilized, and the presence of trees in these areas can help absorb some of the excess nutrients from the fertilizer, preventing it from polluting the water supply (Livesley, McPherson, and Calfapietra 2016; Janke, Finlay, and Hobbie 2017). Planting trees and other plants along waterways is an effective way to catch excess nutrients without disturbing open grass areas such as large, open gathering spaces in urban parks. (Livesley, McPherson, and Calfapietra 2016). Urban trees may participate in a decrease in nutrient deposition onto impervious surfaces and help absorb excessive fertilizer use on grasses close to impervious surfaces (Beck, McHale, and Hess 2016; Janke, Finlay, and Hobbie 2017; Hobbie et al. 2017).

Urban trees are also an effective way to mitigate urban heat islands (Rötzer et al. 2021; Livesley, McPherson, and Calfapietra 2016; Schottland 2019). Trees – especially when they reach mature sizes – provide substantial shade that keeps houses and buildings cooler. Trees intercept 25-90% of short-wave radiation, creating up to a 40-degree difference in temperature compared to unshaded, paved surroundings. Targeted efforts to ensure shading provided by trees has a consequential impact on the urban heat island effect because asphalt and concrete store more energy from the sun and radiate more heat than turf grass often found adjacent to streets (Livesley, McPherson, and Calfapietra 2016; Rötzer et al. 2021). Their effect is the strongest of east-west streets, but it is still significant on north-south streets as well (Livesley, McPherson, and Calfapietra 2016). Canopy density provides an important role in the intensity of heat reduction, with higher-density canopies having a much more significant impact on heat (Rötzer et al. 2021; Livesley, McPherson, and Calfapietra 2016). Urban trees also provide cooling via evapotranspiration. Trees with higher carbon sequestration have more active stomata, so they also have higher rates of evapotranspiration. Enough trees throughout a city can lower the city's overall temperature by several degrees through the effects of shade and evapotranspiration, and cities that prioritize trees with higher carbon sequestration can receive the benefits of reduced temperatures in addition to contributing more atmospheric carbon reduction (Rötzer et al. 2021; Brack 2002; Livesley, McPherson, and Calfapietra 2016; Schottland 2019).

Urban trees can also reduce air pollution, usually by offsetting CO_2 emissions from vehicles in cities. Trees absorb CO_2 via their stomata, and they also capture other pollutants via dry deposition on leaf surfaces (Rötzer et al. 2021; Livesley, McPherson, and Calfapietra 2016). However, some tree species can emit biogenic volatile organic compounds (BVOCs), which are precursors to smog and ozone. It is important to avoid planting tree species that emit BVOCs in an urban setting because they could make air quality worse (Livesley, McPherson, and Calfapietra 2016).

While the benefits of trees in an urban setting are often discussed on a city-wide scale, their direct benefits to city residents can often be dictated by their placement throughout the city. Therefore, efforts to expand and sustain urban forests must consider equity in tree placement to ensure that all parts of the city are equally receiving benefits. Additionally, equitable distribution of urban trees translates to more equitable mitigation to increasing heat due to climate change (Elmqvist et al. 2019; Hunter, Cleary, and Braubach 2019).

Equity in climate action is challenging since many goals can be difficult to directly quantify and progress, but the urban heat island effect is a more-clearly measurable issue that can be mitigated by increased shade and transpiration due to trees (Angelo et al. 2022; Hsu et al. 2021; Ziter et al. 2019). Common focus areas in city climate action plans include focuses on open space with trees and health improvements (Angelo et al. 2022) – two factors that are often improved by the ecosystem services provided by trees. These focus areas appear relatively more in climate action plans and are more clearly targeted by climate action plans because urban trees are a clear existing solution to addressing climate impacts that have clear heat, health, and water quality benefits (Ziter et al. 2019; Angelo et al. 2022; Hunter, Cleary, and Braubach 2019). Communities with less tree equity will feel the heat effects of climate change more than communities with better tree coverage (Angelo et al. 2022), and resources like Tree Equity Explorer (American Forests and EarthDefine 2020) and Vibrant Cities Lab (U.S. Forest Service, American Forests, and National Association of Regional Councils n.d.) are helpful resources for cities looking to improve equity in their urban forests and citizens looking to become more informed.

A common technique for determining the locations of trees is in an area is to use airborne LiDAR to detect tree crowns based on the elevation of the returned LiDAR points (Tanhuanpää et al. 2019; Popescu and Wynne 2004; Klobucar, Sang, and Randrup 2021; Rodríguez-Puerta et al. 2022).

Light Detection And Ranging, or LiDAR, is method to detect the locations and distances of objects with laser pulses (Dong and Chen 2017b). Outside of recent approaches of using LiDAR to detect trees, it is also commonly used for tasks such as identifying building locations, road pavement surveys, object detection for self-driving vehicles, and watershed delineation (Awrangjeb, Ravanbakhsh, and Fraser 2010; Holgado-Barco et al. 2014; Zhang et al. 2018; Yang et al. 2014). LiDAR requires an active sensor, which means that it sends and receives the laser pulses and requires an energy source for it to operate. A computer connected to the sensor measures how long it takes for a laser pulse to return, and combined with data from the Global Navigation Satellite System (GNSS), it deduces the exact location of where the laser pulse bounced off an object based on the speed of light. Each point where a laser pulse bounced off an object it called a *return*. A single laser pulse can have multiple returns if it partially passes through some objects. In the context of the planet's surface, a collection of returns can be visualized in three dimensions as a collection of floating points that follow the shape of the terrain and the objects sticking out of the surface of the earth, called a *point cloud* (Roussel et al. 2020; National Geospatial Program and United States Geological Survey 2018; Dong and Chen 2017b).

LiDAR data is processed using specialized tools to create three different products: the point cloud, a digital surface model, or a digital elevation model (Lambers 2018; Dong and Chen 2017a). The point cloud data is the most computationally expensive, but it retains exact information for each point in the point cloud, including whether it was the first return or a derivative return and the exact location and elevation. Because the point cloud contains more information than other formats, it can be advantageous to perform filtering and processing on the point cloud before transforming it into other formats. A digital surface model (DSM) is a rasterized version (data is stored as pixels instead of points and vectors) of the point cloud that only considers the highest elevation points. During the rasterization process, data about lower elevation points are lost, and only elevation information is retained for each pixel in the DSM raster. DSMs are helpful when attempting to examine the tallest objects in the area, which makes it particularly helpful for evaluating tree canopies. A digital terrain model (DTM) is a rasterized version of the point cloud that only considers the lowest elevation points. It is useful for establishing the elevation and character of the terrain for an area (Dong and Chen 2017a; National Geospatial Program and United States Geological Survey 2018).

LiDAR can be collected in many different ways, but for the purpose of studying urban tree canopies, airborne LiDAR is most helpful. For airborne LiDAR, the required sensors are mounted to a plane, and the plane flies in lines over the entire area for which the LiDAR data must be collected. Spaceborne LiDAR is not effective because satellites are too far from the surface of the planet to capture returns at high enough resolution. Airborne LiDAR can be collected at any time of the year, with certain seasons being more helpful for different purposes (National Geospatial Program and United States Geological Survey 2018; Roussel et al. 2020; Münzinger, Prechtel, and Behnisch 2022). The City of Greenville collects LiDAR every winter because it is most ideal for city planning purposes ("GIS Data Access" n.d.).

Through its various initiatives and ordinances to support and expand green spaces, the City of Greenville in South Carolina, United States has proven its interest in Green Infrastructure, which includes a significant emphasis on the many benefits provided by trees. The majority of Greenville's trees were planted by the Civilian Conservation Corps in the 1930s, and in recent years, Greenville has created commitments to expand tree coverage before those trees die (Greenville, SC 2021). Through their TreesGVL program, Greenville has committed to planting 1000 trees each year in public parks and rights-of-way, funded by the City Tree Fund Foundation. As part of the TreesGVL program, the city has committed to conduct a multi-year public awareness campaign on the many benefits of trees, which will likely help inform people of the value of trees in an urban landscape outside of air quality and shade (Drillet et al. 2020; "Greenville's Trees" n.d.). Greenville also recently passed a tree ordinance that adds restrictions for the removal of trees, including special restrictions and fines for the removal of large trees that are classified as "heritage trees" ("Ordinance No. 2021-07" 2021).

However, Greenville does not have a comprehensive database of trees (Greenville, SC, PlanIT Geo, and TreePlotter n.d.), making it difficult to detect areas in need of trees, track changes in tree placement over time, enforce the tree ordinance, and compute tree benefits for the city. Further, without a comprehensive database of tree locations and sizes, it is difficult to quantify the ecosystem services provided by the urban forest. Many cities – including the City of Greenville in South Carolina, Untied States – are interested in receiving and computing the benefits of trees, but they lack city-wide knowledge of where trees are located. Greenville is currently working to incrementally create a database of trees using the TreePlotter software (Greenville, SC 2021; Greenville, SC, PlanIT Geo, and TreePlotter n.d.), but it requires manual entry by city staff members and could be supplemented by automatically created city-wide database of tree locations and approximate sizes in order to make it easier to identify locations that could benefit from the addition of trees.

There is clearly a need for an alternative way to develop a database of tree locations in the City of Greenville that does not require slow, manual identification and logging of tree data. Since the City of Greenville and various federal agencies collect detailed remote sensing data, it is possible to discern the locations of trees by processing available data using GIS techniques. The goal of this project is to develop an automated process and algorithm for the City of Greenville to map the locations of trees along city streets to help and supplement the city's tree planting and enforcement goals. The results will provide critical data for computing the benefits of trees and evaluating tree distribution equity, and the process can be applied to other cities in the region.

2. Methods

The ForestTools R package (Plowright [2016] 2022) was used to detect the locations of trees using a digital surface model (DSM) generated from a LiDAR point cloud. Before generating the DSM, the LidR R package (Roussel et al. 2020) was used to remove buildings, unvegetated areas, and planar points from the point cloud for improved detection accuracy. Vegetated vs unvegetated areas were determined using multispectral photography. The primary area of concern is the space between streets and buildings since that is typically viewed as public or shared space, so at every stage, processing was constrained to a 25 meter buffer from the pavement edge of the city streets. The general workflow for detecting trees is visualized in Figure 1 followed by a detailed description within each step.

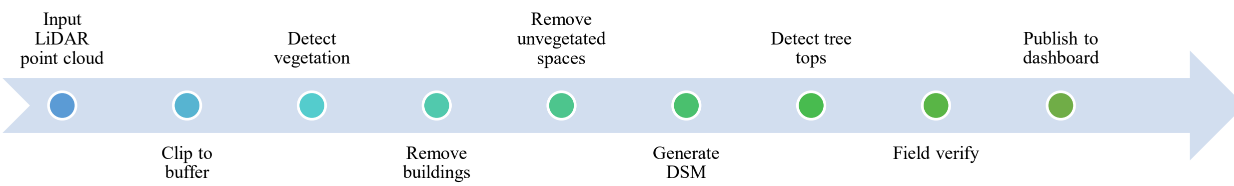


Figure 1. Workflow diagram for detecting the locations of trees.

2.1. Vegetation region identification

The vegetated areas were detected with the normalized different vegetation index (NDVI). NDVI creates a raster with pixel values representing the relative amount of photosynthetically active vegetation cover. Values of 1 represent the highest amount of photosynthetically active vegetation, and values of -1 represent the least amount of photosynthetically active vegetation. It was calculated using near-infrared (NIR) and red bands from multispectral imagery of South Carolina using the relationship defined by Tarpley et al. (1984):

$$\text{reflectance} = \frac{\text{NIR} - \text{red}}{\text{NIR} + \text{red}} \tag{1}$$

Positive pixel values represent pixels with a relatively higher density of photosynthetically active biomass and were interpreted as pixels with vegetation. The multispectral imagery of South

Carolina was acquired from the National Agricultural Imagery Program on June 27, 2022. (NAIP 2021)

A new raster representing all vegetated regions was created by re-coding all pixels with NDVI values greater than 0.1 to 1 and other values to 0. The threshold was set to 0.1 to ensure that nearly all vegetation was included while also excluding some objects that are misinterpreted as vegetation. Patches of some flat building tops received NDVI values between 0 and 0.1, perhaps due to mosses growing on their roofs. To reduce noise, groups of three or less pixels were pruned from the raster. The raster was converted to unsmoothed polygons for use in special merge operations on the LiDAR point cloud.

2.2. LiDAR cleanup

Buildings and other tall structures were removed from the LiDAR point cloud to remove objects that could be confused by the tree detection algorithm as tall trees. The algorithm expects trees to be the tallest objects. The latest available LiDAR and building data for Greenville County at the start of this project were from Winter 2020.

A double spatial merge between the LiDAR point cloud and the building polygons was performed to identify the three-dimensional points that represent buildings. The first spatial merge added a Boolean attribute stating whether the LiDAR point was within the building footprint, and the second spatial merge added an attribute containing the height of the building. The building height attribute was increased by 1.5 meters to encapsulate rooftop structures such as HVAC units or chimneys. Points were removed if they were in a building footprint and their elevation was lower than the building height attribute using LidR’s `ẽħƙŝðõ¿ɔɳħ` function, which allows filtering based on attribute. Buildings were removed first because their removal resulted in a substantial reduction of points in the point cloud, which also allowed subsequent steps to process through the point cloud faster.

Unvegetated areas, previously determined using NDVI, were removed from the LiDAR point cloud with a similar process. A spatial merge was performed between the vegetation polygons and the LiDAR, and the `ẽħƙŝðõ¿ɔɳħ` function was used to remove points that are not within the vegetation polygon.

Though buildings, roads, bridges, and other unvegetated objects were usually removed from the previous steps, not all objects were always filtered from the point cloud. To eliminate the chance of misinterpreting those objects as trees, planar points were removed from the LiDAR point cloud. This does not have an effect on points representing trees because the tree canopies do not follow a planar pattern. Planar points were identified using a combination of the LidR package’s `rðəlðḷŝ¿rġɢɔðr` and `rġɔ¿ġɔkɢḷð` functions to add a Boolean plane attribute to each point in the LiDAR point cloud. The `ẽħkŝðõ¿ɔṇħ` function was then used to remove planar points from the LiDAR point cloud.

2.3. Digital surface model

A digital surface model (DSM) was generated from the filtered LiDAR point cloud. The digital surface model is a raster dataset that represents the highest elevation surface above the ground (as opposed to a digital terrain model, which represents the ground elevation). In the context of visualizing a city's tree canopy, a DSM dataset is created by selectively removing point data until only data for the tree canopy remain, and then the highest elevation points of the tree canopy are retained in the DSM. Working with a DSM to identify tree locations is significantly faster than working directly with the point cloud since a DSM stores less data. Each pixel in the DSM raster only contains an elevation value. If multiple points from the LiDAR point cloud corresponded to the same pixel in the DSM raster, the point with the highest elevation was used. The DSM was generated with a cell size of 2 ft^2. The LiDAR data from Greenville County were from the winter, and most trees in the city are deciduous, so there were no leaves on most trees in the point cloud. Therefore, once non-tree points were removed, the resultant LiDAR point density was very low. To ensure that the algorithm had enough data to deduce the locations of trees, voids in the DSM dataset were filled using a natural neighbor algorithm that made a best guess at appropriate pixel values when points were missing based on adjacent pixel values. Sometimes, a DSM may have pixel values normalized to represent heights above ground instead of raw elevations, called a canopy height model (CHM). The DSM was not normalized to a CHM because the computational and time requirements of converting a large DSM to a CHM were not worthwhile since the terrain in the City of Greenville lacks substantial elevation changes that could affect the interpretation of trees. Conversion to a CHM may be needed in cities with steeper hills.

2.4. Tree top detection

The variable window filter (VWF) algorithm provided by the ForestTools R package was used to detect the locations of trees by evaluating regions of local elevation maxima from pixel values in the DSM. Each local maximum was individually evaluated by applying the variable window filter in a circular shape around the local maximum. Each detected elevation maximum in the DSM was assumed to approximate the center point of the tree.

The VWF algorithm was configured to exclude DSM pixels with an elevation of less than two feet to only attempt to detect trees at least two feet tall. The ideal variable window filter for mixed forests determined by Popescu and Wynne (2004), $crown\ width = 0.00901x + 2.51513$, was used.

2.5. Publishing and verification

The detected tree locations were placed on a map along with satellite imagery of the City of Greenville. Empty attributes were added to each detected tree for the species name, diameter at breast height (DBH), and whether the tree has been verified in the field. A 100-meter fishnet grid was generated for the entire City of Greenville so individual grid cells could be isolated for accuracy verification. The project was published on ArcGIS Online as a service layer so it can be used with other web and mobile based apps.

Verification of trees was primarily performed by cross-analysis of Google Street View, high-resolution winter aerial images, and multispectral imagery. Many residential backyards were verified without field verification via Google Street View images. In patches of dense growth, verification was skipped since individual trees could not be reliably distinguished. For select locations, the ArcGIS Collector and Field Maps apps were used to field verify trees along streets. These apps were primarily used along streets with high spacing between trees since mobile phone GPS precision tends to be too low to distinguish closer trees. Trees in fenced-off areas were not field verified to avoid issues with trespassing on private property.

For the purpose of examining accuracy: local maxima that were correctly detected as treetops are considered true positives, and local maxima that were incorrectly identified as trees are considered false positives. Some trees were not detected as treetops because local maxima were not distinguished from the digital surface modal, and these are considered as false negatives. The distinction between false positives (wrongly detected trees) and false negatives (missed trees) provides an indicator for whether the detection algorithm tends to overcount or undercount potential trees.

3. Results

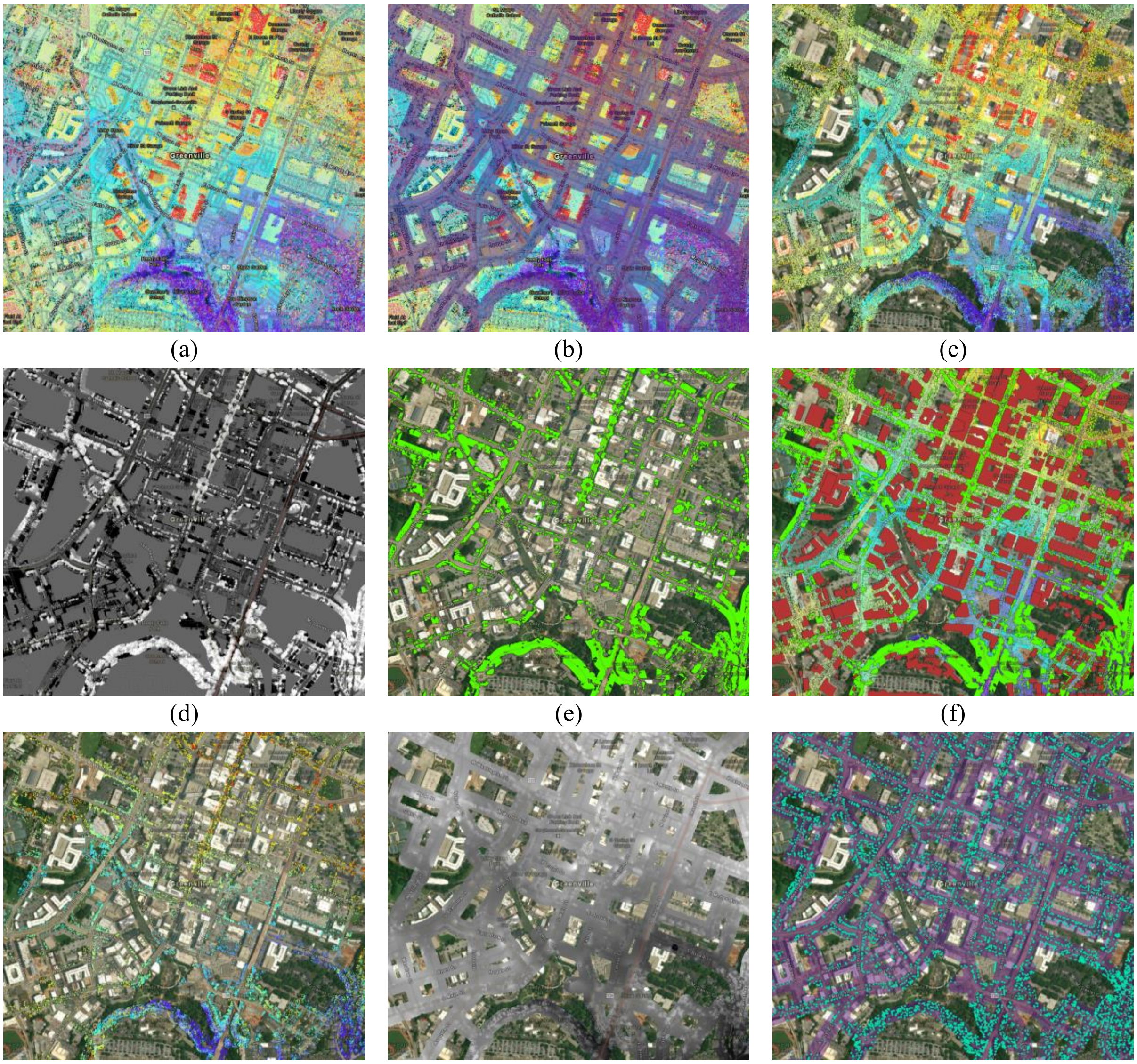
3.1. Vegetation regions

The vegetation detection process revealed the locations of vegetation along the streets within the 25-meter buffer (see Figure 2g and Figure 2h). The road buffer region has an area of 34.44 km^2 (13.30 mi^2) and the vegetated regions within the road buffer covered 17.21 km^2 (6.64 mi^2). Therefore, 49.97 percent of the road buffer was considered to have detected vegetation. The difference between trees, bushes, and grasses could not be confidently distinguished in all cases by NDVI, so this 49.97 percent coverage statistic includes all vegetation – not just trees.

The vegetation detection process also yielded information on estimated tree coverage for the entire city. When tree coverage was investigated, a higher NDVI threshold was used, which made it possible to exclude most grassy areas when assessing tree coverage for the entire city. This tree coverage was compared to zoning data for the city (see appendix section 7.3 for zoning data processing), proximity to the center of roads, and density of tree coverage. Tree coverage was also compared against detected tree heights.

Overall, 22.07 percent of the area is covered by trees or other vegetation. Single-family residential districts had the highest percent vegetation cover at 31.90 percent, followed by single-family and multifamily residential districts at 25.66 percent, neighborhood commercial districts at 24.72 percent, office districts at 22.41 percent, and planned development districts 18.74 percent (Figure 3). Multifamily had the least percent tree coverage at 3.73 percent.

The percentage share of vegetation across zones was also calculated. It revealed that single-family residential zones contain the most vegetation coverage in the city limits (Figure 4). Single-familiy residential zones are the most common zone type in the city (Figure 5).



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Figure 2. A visualization of each major stage of preprocessing and processing for the tree detection process, centered over downtown Greenville. (a) The LiDAR point cloud over downtown Greenville, SC. (b) Visualization of the 25 m street buffer. (c) The LiDAR point cloud after it was clipped to the street buffer. (d) The results of the normalized difference vegetation index. The lighter areas represent high reflectivity and darker areas represent low reflectivity. Areas outside of the street buffer are middle gray. (e) A visualization of the vegetation polygons generated from the NDVI. The light green areas represent vegetated areas within the buffer. (f) A visualization of the vegetated areas and buildings that were used to filter the LiDAR point cloud. Vegetated areas are green. Buildings are red. (g) The remaining points in the LiDAR point cloud after the buildings and unvegetated areas are removed. (h) The digital surface model generated from the filtered LiDAR point cloud. It encompasses the entire buffer due to void filling. (i) The detected tree locations are represented by turquoise dots.

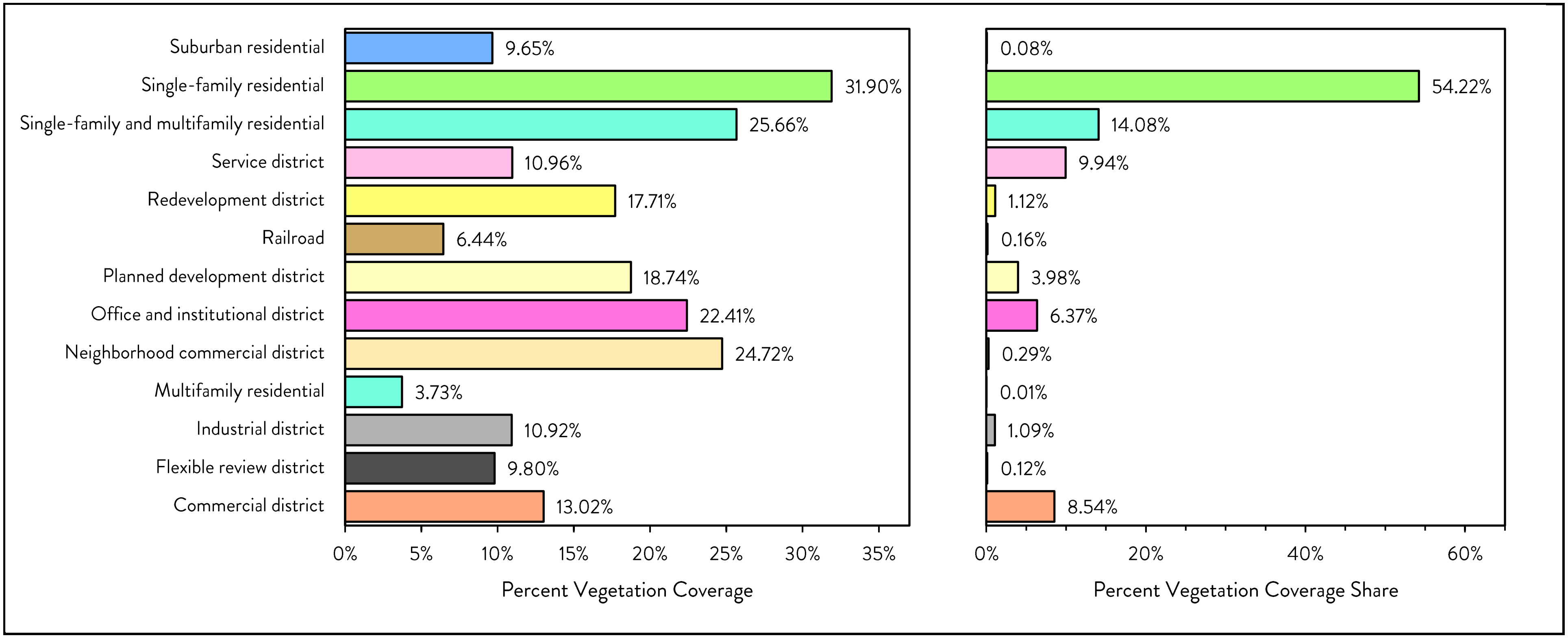


Figure 3. Percent coverage of trees along the 25-meter street buffer for grouped classified zones in the City of Greenville. Classified zones contain multiple subgroups that are not distinguished on this chart.

Figure 4. Percent share of vegetation coverage for grouped classified zones in the City of Greenville. Percentages sum to 100%.

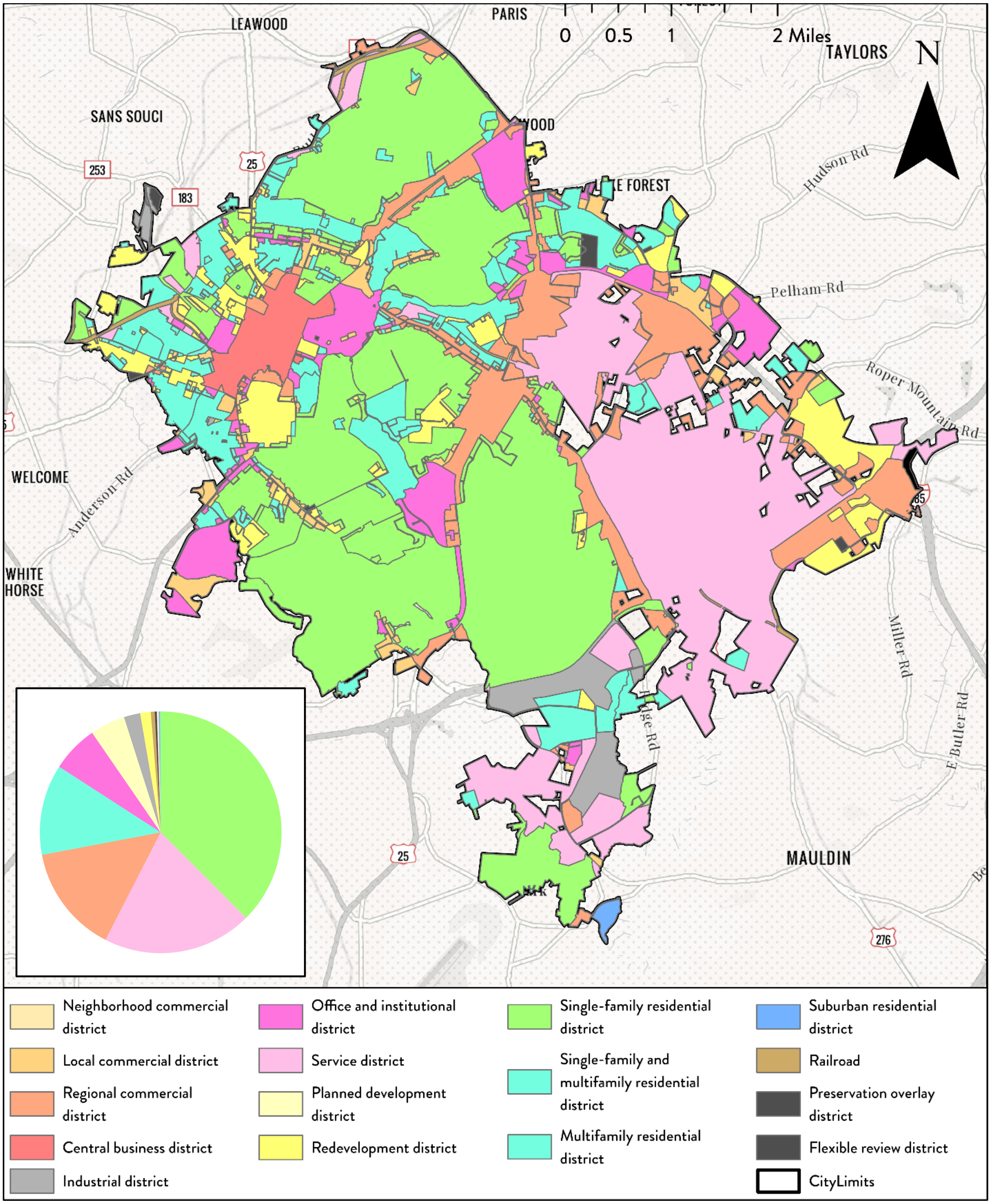


Figure 5. Classified zones overlayed on a map of Greenville, SC. The proportion of zone coverage in the City of Greenville is shown in the inset pie chart. Single-family residential districts and service districts have the most area in the city, and multifamily residential districts have the least area in the city. *Data from City of Greenville, Greenville County, Esri, HERE, Garmin, SafeGraph, Geo Technologies Inc, METI/NASA, USGS, EPA, NPS, and USDA.*

In general, vegetation coverage (Figure 6) increased further from the streets (into the property), and it was less prevalent along the streets in most places except areas like Main Street. Vegetation was considered to be along streets if it is less than 2 meters from the edge of the street, very close to streets if it was from 2 meters (inclusive) to 12 meters (exclusive), and further from the streets if it was from 12 meters (inclusive) to 25 meters (inclusive).

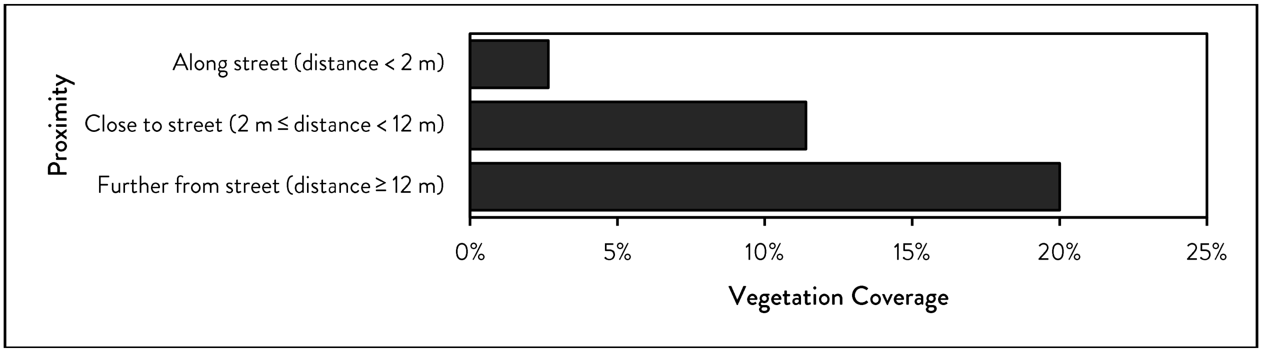


Figure 6. Percent coverage of tree-sized vegetation based on different proximity classifications. Classified zones contain multiple subgroups that are not distinguished on this chart.

3.2. Tree detection

A total of 42,074 trees were detected within twenty-five meters of streets in the City of Greenville. Trees were detected using a multi-stage process to detect and isolate vegetated areas of the city and detect individual trees tops, which is visualized in Figure 2.

The detection process included removing extraneous points from the LiDAR point cloud. Before any filtering of the LiDAR point cloud, there were 48,627,503 points contained within the 25-meter street buffer. After the LiDAR point cloud was filtered – buildings were removed to eliminate tall points that do not represent trees, unvegetated regions were removed since unvegetated regions do not contain trees, and planar points were removed to help eliminate fences, power lines, and walls – only 17,330,935 points remained in the point cloud (35.64 percent of original points). The reduction of points is visualized by Figure 2c, Figure 2f, and Figure 2g for downtown Greenville. The removal of extraneous points from the LiDAR point cloud improved the ability for the detection algorithm to determine tree locations.

The tree detection results were verified by randomly sampling 100 m^2 areas of the city for manual verification (Table 1). During the verification process, there was no distinction made between tree species and size.

Investigation of the points identified as trees indicated that not all of them actually represent individual trees. 33 percent of verified tree points were determined to be wrongly detected as trees due to overcounting large trees or detecting objects that were not filtered from the LiDAR point cloud as trees. 14 percent of verified trees points were missed by the detection process due to trees that were obscured by other taller objects, such as taller trees or building overhangs that were not removed from the LiDAR point cloud during the filtering process.

The accuracy of the detection process is expressed in terms of detection success and detection correctness. Detection success is a measure of the percentage of trees that were detected by the detection process. In the randomly selected sample area, 79 percent of trees were successfully detected. Detection correctness is a measure of the percentage of detected points that correspond to trees. In the randomly selected sample area, detected tree points were accurate 53 percent of the time.

Table 1. Accuracy information for the tree detection process. False positives usually occur when a single tree on the ground is detected as more than one tree by the model. False negatives occur when a tree is missed by the detection process. True positives represent correctly detected trees.

Outcome	Meaning	Count	Percent
True Positives	Correctly detected trees	408	53 %
False Positives	Wrongly detected trees (overcounted or not trees)	256	33 %
False Negatives	Missed trees	106	14 %

A clear example of the accuracy of the tree detection process can be seen at the intersection of McBee Street and Main Street in Downtown Greenville (Figure 7). The eight large deciduous trees at the intersection of the two streets are detected as 30 distinct trees. The smaller deciduous trees near the intersection are accurately detected.

Figure 7. Detected trees at and near the intersection of Main Street and McBee Street in downtown Greenville. Pink areas represent vegetation and dots represent local maxima detected as individual trees. *Basemap from National Agricultural Imagery Program (NAIP 2021).*

Since the accuracy assessment relied on verification of detected tree points from a random sample of areas within the City of Greenville, the results of the accuracy assessment can be extrapolated to apply to the entire City of Greenville.

3.3. Tree distribution

Results from trees density analysis show that trees were not evenly distributed throughout the city (Figure 8; Figure 9). Trees were concentrated in residential areas and designated parks.

Comparison of mean tree height and percent vegetation cover for 25 m^2 cells along the street buffer revealed that percent vegetation cover was low around the street buffer, but most instances of vegetation cover on the higher end only occurred when mean tree height was also high (Figure 10).

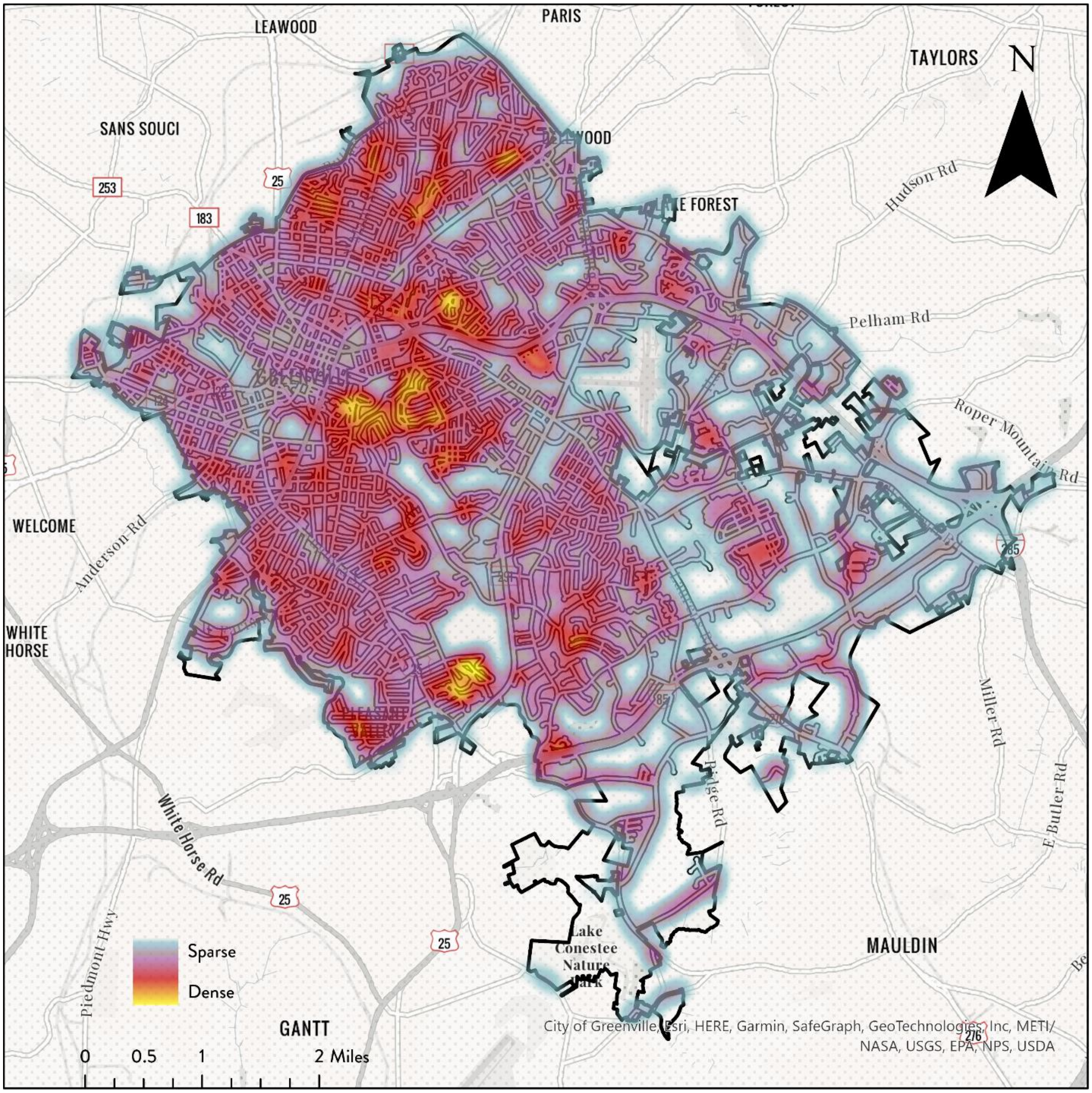


Figure 8. Heat map of detected trees within 25-meter street buffer. Blue is low density, red is medium density, and yellow is high density. The area of highest density near downtown Greenville is Cleveland Park. *Basemap and boundary data from City of Greenville, Greenville County, Esri, HERE, Garmin, SafeGraph, Geo Technologies Inc, METI/NASA, USGS, EPA, NPS, and USDA.*

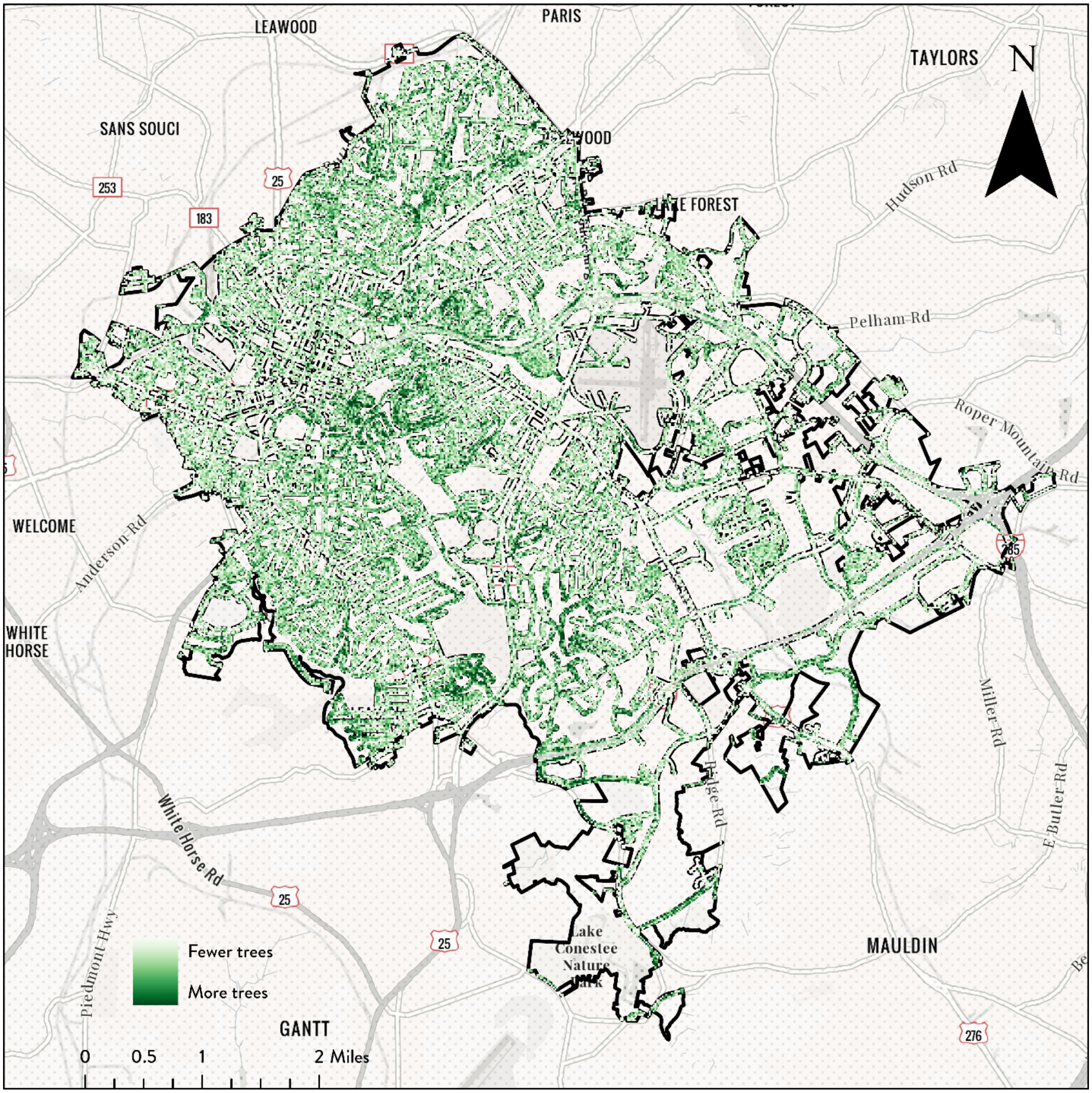


Figure 9. Choropleth map showing tree counts within 25-meter street buffer. Detected tree points are aggregated into 25 m^2 cells. Cells with more detected trees are shaded darker green, and cells with fewer trees are shaded lighter green. *Basemap and boundary data from City of Greenville, Greenville County, Esri, HERE, Garmin, SafeGraph, Geo Technologies Inc, METI/NASA, USGS, EPA, NPS, and USDA.*

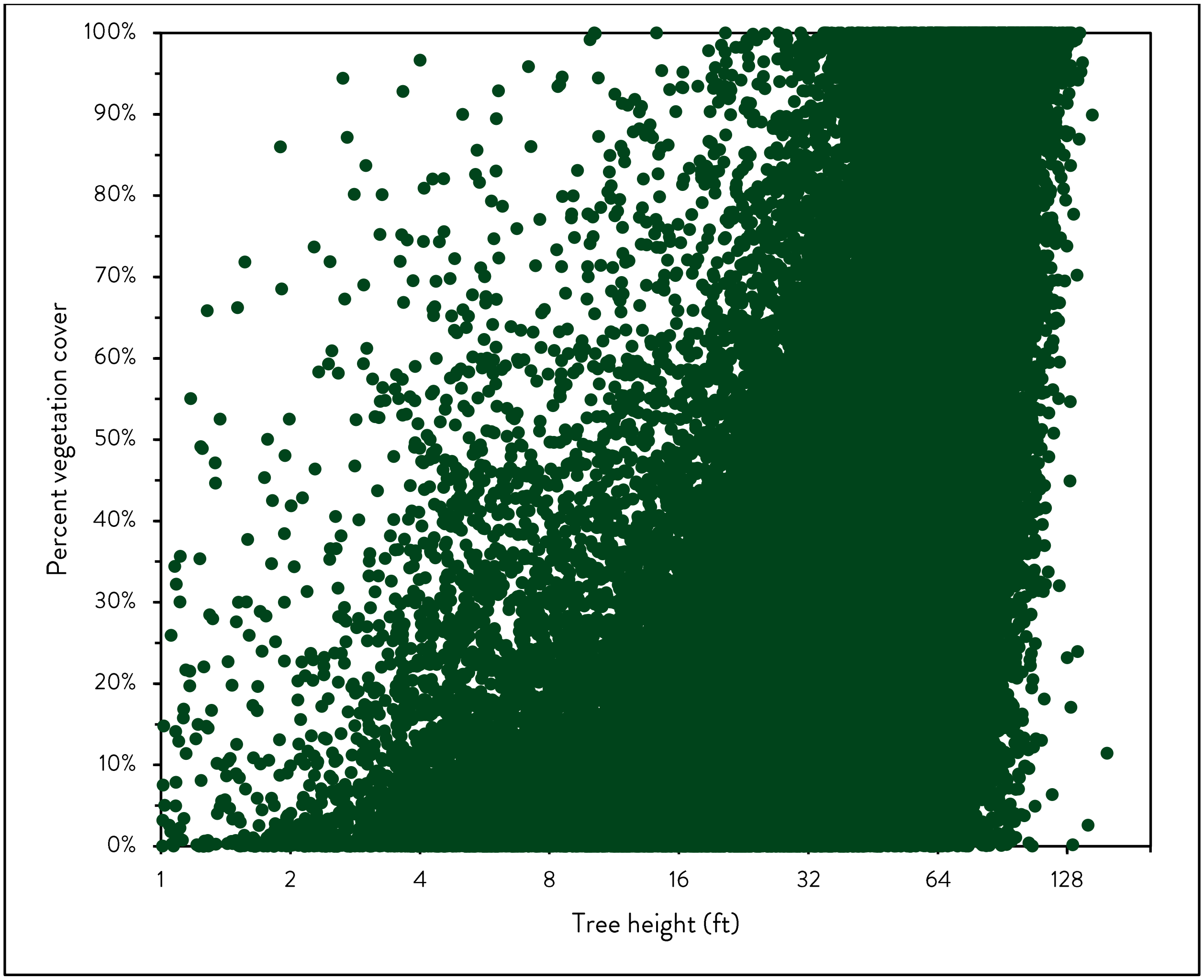


Figure 10. Correlation between tree coverage percent and average tree height for 25 m^2 cells. Each cell represents an aggregation of average tree heights in a 25 m^2 area. Cells were only created along a 25 m street buffer.

3.4. Census connections

Tree coverage and heights were also evaluated through the lens of census block groups from the 2020 decennial census and Greenville's designated special emphasis neighborhoods (Figure 11). The City of Greenville contains 79 census block groups and 13 special emphasis neighborhoods, and for the purpose of analysis, census block groups that cross the city boundary were clipped to the extent of the city boundary.

For each census block group, choropleth maps were created to visualize tree coverage in comparison to income and age indicators (Figure 12). Tree coverage was computed for all areas within 25 meters of the edge of streets. The census block groups in the city with the lowest tree coverage group (0.0-1.2 percent) were the eastern and western block groups within the central horizontal band of block groups. Many of these census block groups correspond to Greenville's special emphasis neighborhoods. The census block groups that fit into the highest tree coverage group (4.9-7.7 percent) were at the north of the city, near Cleaveland Park (southeast of downtown), and near Conestee Nature Preserve (in the south). The remaining block groups were generally placed into the 3.0-4.8 percent coverage group.

Additionally, for each census block group, the percentage of tree coverage and average tree height were compared to median age, median income, household and family size, highest level of educational attainment, and employment status via scatterplots (Figure 13). Each comparison demonstrated a slight trend via a regression line, but no comparison had a strong R-squared value.

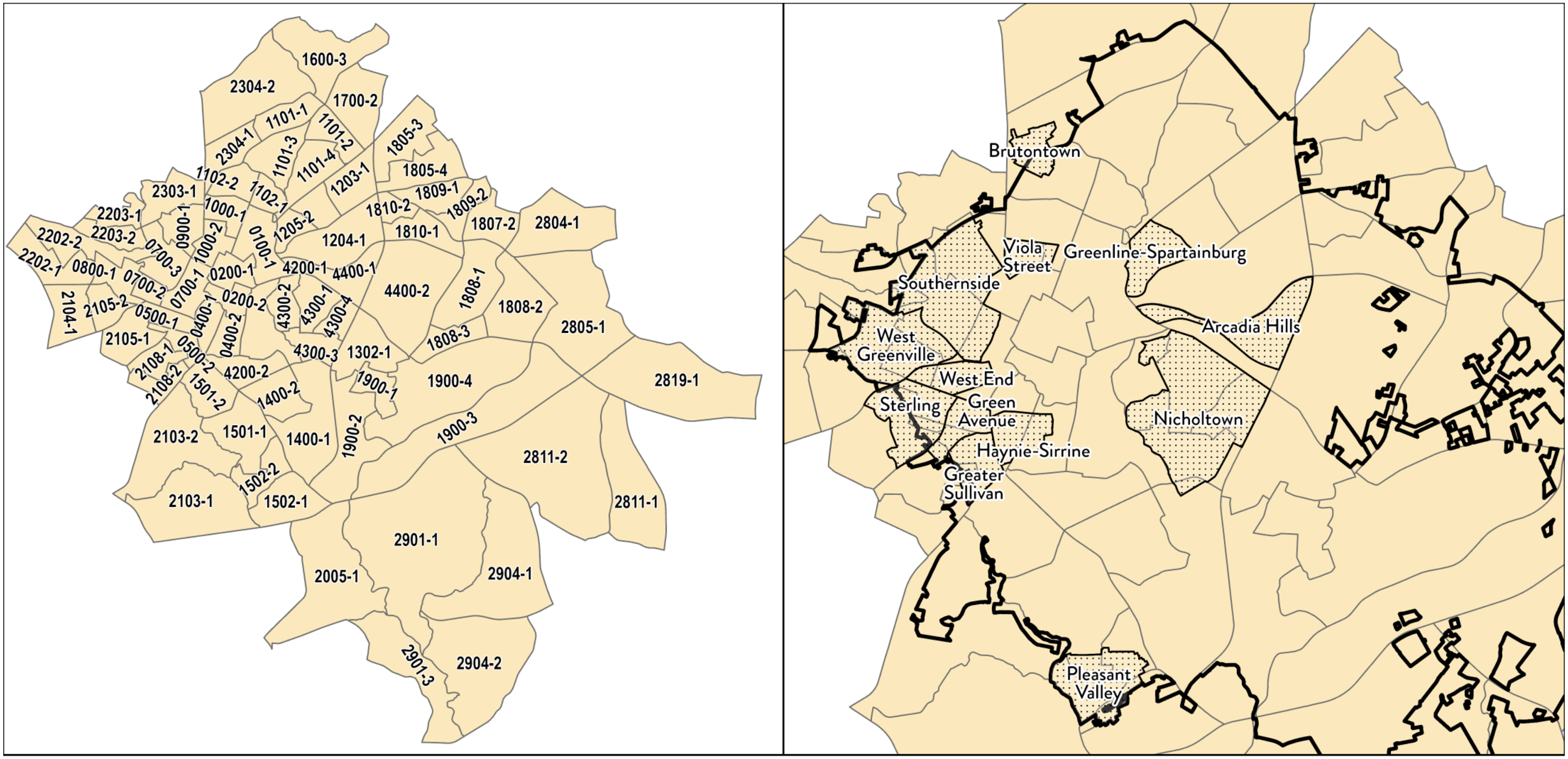
For the scatterplot of unemployment and percent tree vegetation coverage, only civilians aged 16 or older were considered for unemployment status due to the way census data is reported. The data demonstrated a slightly negative trend (less unemployment with more tree vegetation coverage), but no significant correlation can be discerned. The regression line had an R-squared value of 0.0024.

For the scatterplot of average household income and percent tree vegetation coverage, the data revealed a slightly positive trend, and most points were visually near the regression line. The regression line revealed an R-squared value of 0.008, so no statistically significant correlation could be reported.

For the scatterplot of median age and percent tree vegetation coverage, median age was calculated for the entire census block group since higher specificity data were not available. The regression line revealed a slightly positive trend, but the R-squared value was only 0.001.

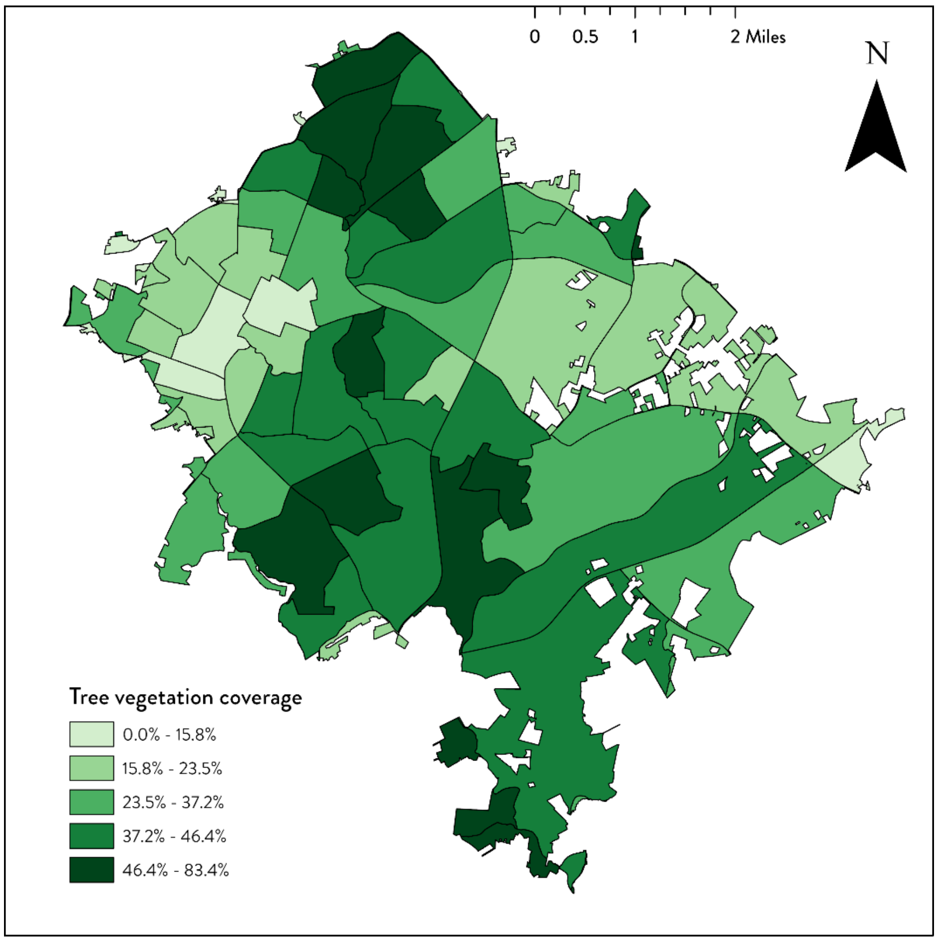
The scatterplot of average household size and percent tree vegetation coverage revealed a slightly positive regression line but no discernable correlation. The R-squared value was only 0.0116.

For the scatterplot of the percentage of the population in the census block group who have attained a college degree and percent tree vegetation coverage, only civilians aged 25 or older were considered to account for the time it takes for a college degree to be earned. The data demonstrated a positive trend (higher college-educated block groups tended to have more tree vegetation coverage), but no statistically significant correlation can be discerned. The regression line only had an R-squared value of 0.0218.

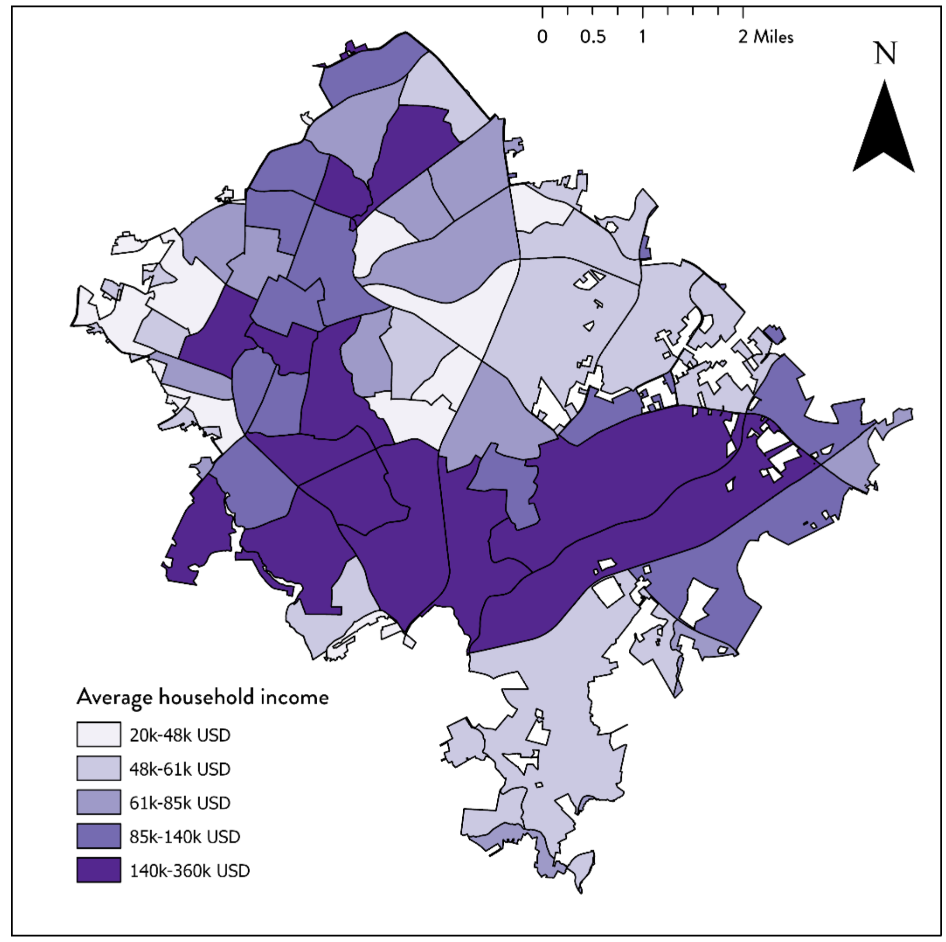


(a) (b)

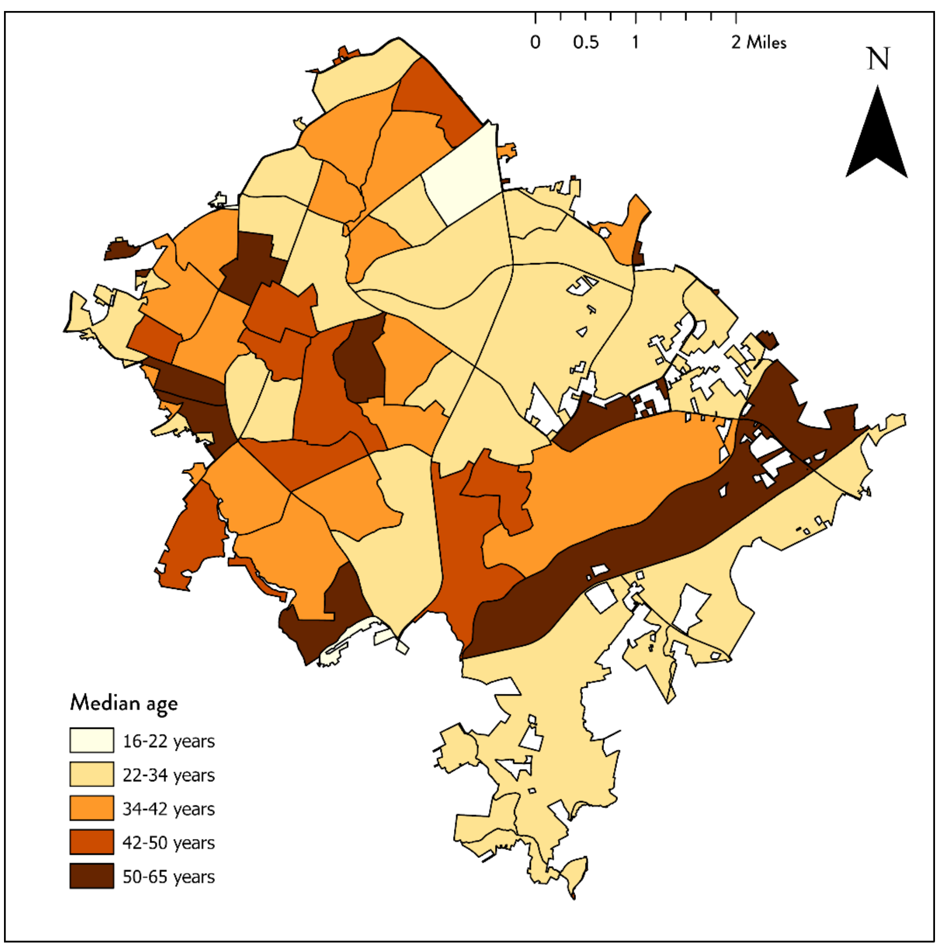
Figure 11. Labeled census block groups in the City of Greenville compared with the city's special emphasis neighborhoods. (a) Census block groups for the City of Greenville. Labels consist of the census tract number followed by the census block group number separated by a hyphen. There are 79 different census block groups in the City of Greenville, but many of the block groups near the border for the city are not exclusively in the City of Greenville. (b) Visualization of special emphasis neighborhoods in Greenville. Special emphasis neighborhoods in are located closer to downtown Greenville and are historically Black communities where over half of households earn less than 80 percent of the area's median income. There are 13 designated special emphasis neighborhoods. Many of these neighborhoods extend outside the city limits, which is denoted by the bold black line. *Census data from NHGIS (Manson, Steven et al. 2022). City boundary and special emphasis neighborhoods from the City of Greenville GIS Division ("GIS Data" n.d.).*



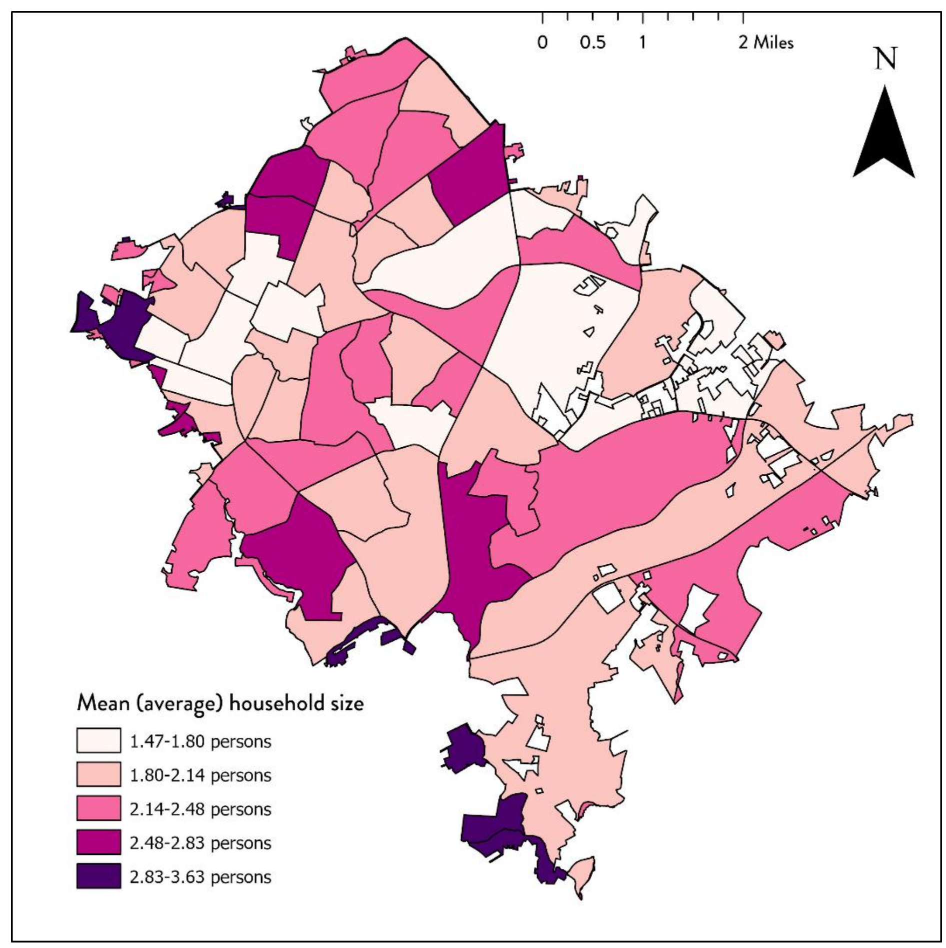
(a)



(b)



(c)



(d)

Figure 12. Choropleth maps showing relevant census block group data for the City of Greenville. (a) Percent tree vegetation coverage for each census block group. Darker shading indicates more coverage. (b) Average household income for each census block group. Average household income is the sum of all income in the census block group divided by the number of households. Darker shading indicates higher average household income. (c) Median age of residents for each census block group. Darker shading indicates higher median age. (d) Mean household size for each census block group. Averages close to two are most common. Darker shading indicates a higher average household size. *Census data from NHGIS (Manson, Steven et al. 2022)*.

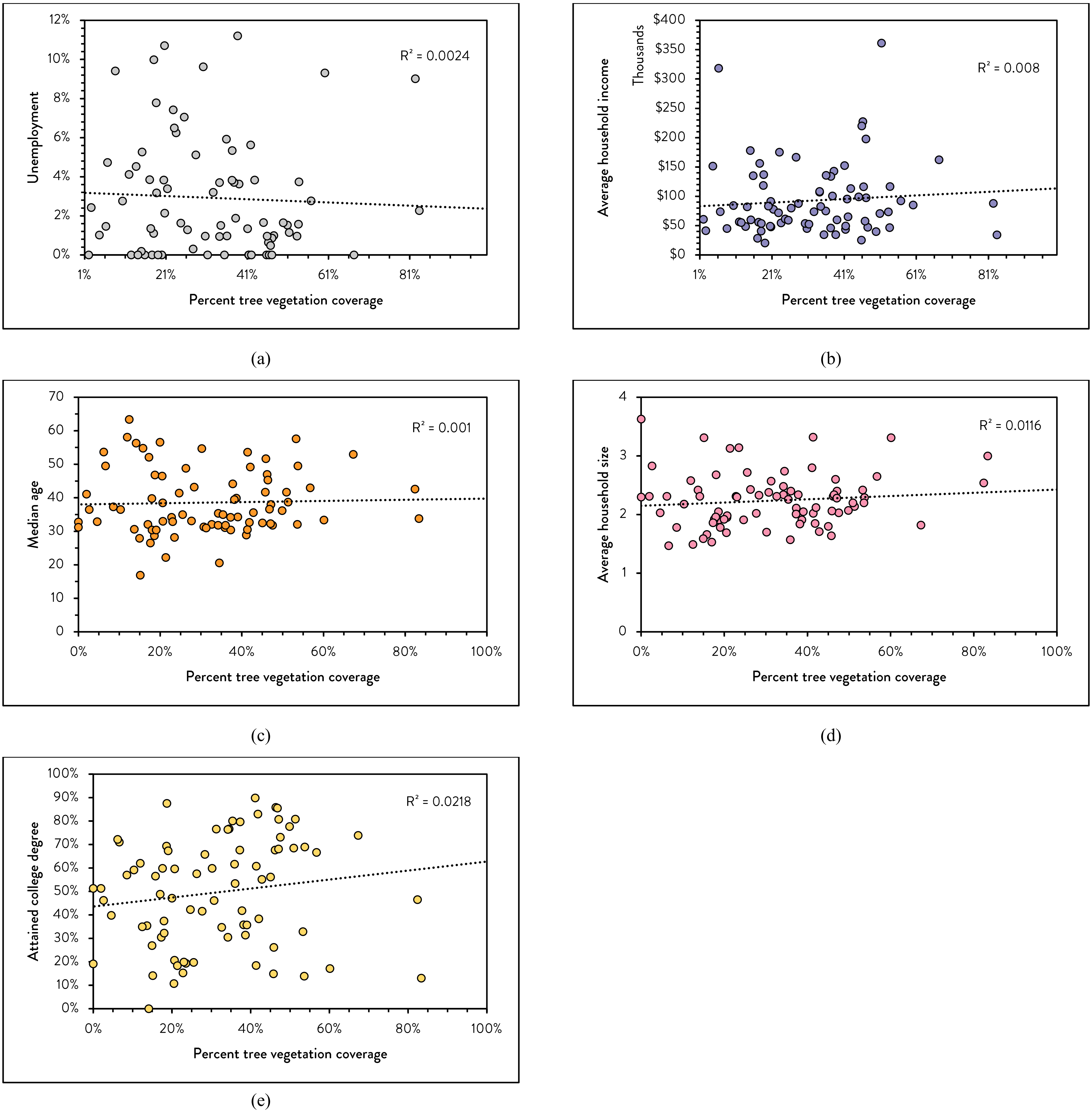


Figure 13. Scatterplots showing relationships between census block group data and tree coverage. All examined relationships had low R-squared values.

Table 2. Classified tree coverage for each census block group in Greenville. Census block groups are distinguished with the census tract number and census block group number separated by a hyphen.

Better tree coverage coverage ≥ 37.2%		Moderate tree coverage 37.2% > coverage ≥ 15.8%		Poor tree coverage coverage < 15.8%
0400-2	1809-2	0100-1	1809-1	0200-1
1101-1	1900-1	0200-2	1810-1	0500-1
1101-2	1900-2	0400-1	1810-2	**0700-1**
1101-3	1900-3	0500-2	**1900-4**	1600-3
1101-4	2005-1	0700-2	2103-2	1700-2
1102-1	2103-1	0700-3	2105-1	1805-3
1102-2	2202-1	0800-1	2108-1	1805-4
1204-1	2203-1	0900-1	2108-2	2104-1
1205-1	2901-1	1000-1	2202-2	2105-2
1205-2	2901-3	1000-2	2304-2	2203-2
1302-1	2904-2	1203-1	2805-1	2303-1
1400-1	4200-1	1501-2	2811-2	2304-1
1400-2	4200-2	1502-1	2904-1	2804-1
1501-1	4300-1	1808-1	4300-4	2811-1
1502-2	4300-2	1808-2	4400-1	2819-1
1807-2	4300-3	1808-3	4400-2	

Each census block group was also classified into tree coverage groups based on poor, moderate, or better tree coverage percentages (**Table 2**). Block group 1 in census tract 2005 had the most tree coverage, and block group 1 in census tract 2901 had the least tree coverage. Better tree coverage was classified as at least 37.2 percent coverage and the largest census block group in the better tree coverage category was block group 1 in census tract 1101 near the southern branch of Greenville. Moderate tree coverage was classified as having between 15.8 percent and 37.2 percent coverage, with the largest block group in the moderate category being block group 4 in census tract 1900 in the central southern section of Greenville. Poor tree coverage was classified as having less than 15.8 percent coverage, and the largest census block group with poor coverage was block group 1 in census tract 700 in the western side of Greenville.

4. Discussion

4.1. Vegetation coverage and distribution

Due to the Agricultural Act of 2014, the USDA Forest Service Forest Inventory and Analysis (FIA) program is inventorying urban forests for the entire country by collaborating with municipalities to collect tree data and generate urban forest canopy coverage information. As a result, many municipalities primarily rely on data collected and processed in conjunction with the FIA program for its ease of use through the i-Tree tools and the ability it affords to easily compare urban tree canopies across different municipalities (Edgar et al. 2021; “About I-Tree” n.d.).

i-Tree has emerged as an easy-to-software suite for comparison between municipalities, including between Greenville and nearby municipalities. Within the county, i-Tree reports that the City of Greenville is ranked 10 out of 11 for canopy area, indicating that there are needed improvements to the tree canopy (“Tree Canopy in Greenville, SC” 2023). Understanding the distribution of tree coverage throughout the city can help identify where the City of Greenville is doing well and where it can improve.

Tree coverage was highest in residential areas. This includes single and multi-family residential space. Tree coverage was likely higher in residential areas since many neighborhoods are characterized by spaced-out houses with large amounts of yard space between houses, streets, and backyards. Additionally, compared to other types of zoning (e.g., commercial or industrial), parking lots are nearly nonexistent in residential spaces, which leaves more potential for grasses, shrubs, and trees to grow in residential areas. Residents may also explicitly plant trees in order increase privacy for their home, and they may also plant trees for the shade they provide in the warmer months. Multi-family zoning is often more dense than explicitly single-family zoning, so it unexpectedly had a lower percentage tree coverage than single-family housing zones. However, it still had a high percent tree coverage for similar reasons: there are still fewer paved surfaces than other zones, and residents still have a desire for attractive landscaping that also creates nice outdoor green spaces.

Tree coverage was the lowest in industrial zoning. Industrial zones are often characterized by large buildings and parking lots. This severely restricts the potential for the presence of trees. These areas may have some isolated trees in their parking lots and along the street, but the vast majority of developed commercial property is covered in impervious surfaces.

A clear example of limited tree coverage in industrial zones in Greenville is the industrial property owned by Duke Energy in the southern part of the city. Most of the property is a gravel lot that is used as staging ground. The remaining portion of the property is parking lots, buildings, and a small amount of buffer trees along the edge of the property.

Commercial zones may not seem like areas that would have much tree coverage – they are often characterized by storefronts with extensive parking lots either in front or beside stores – but in the City of Greenville, commercial zones generally contained a moderate amount of tree coverage. Parking lots, sidewalk areas, and buffer areas behind commercial buildings offer plenty of opportunities for tree coverage. Newer developments also follow city regulations that mandate the replanting of trees that are lost during construction and encourage trees throughout parking lots.

When vegetation coverage was evaluated with respect to Greenville's special emphasis neighborhoods, it became visually clear that most special emphasis neighborhoods were within areas of the city with less vegetation coverage. These neighborhoods are neighborhoods designated by the city as needing extra support in order to protect the residents during redevelopment plans. In these neighborhoods, at least half of residents have income levels less than 80 percent of the area's median income (Kolb et al. 2023; Hughes 2023). The ecosystem services provided by urban trees – especially reduction of heat – has a stronger impact on communities of lower income, so the reduced tree coverage in these neighborhoods relative to most of the city is concerning. Expanding tree coverage may have a significant impact on reducing heat in the special emphasis neighborhoods, which could reduce cooling costs and reduce negative health impacts due to prolonged exposure to high levels of heat in the summer. In many of these neighborhoods, residents have also expressed interest in more parks and tree maintenance/upkeep (Brewer et al. 2015; Rutherford et al. 2016; "Neighborhood Plans" n.d.). These neighborhoods need extra support to increase their tree coverage without displacing current residents and sacrificing safety.

With the recent tree ordinance, areas with fewer trees may begin to see a reduction in the rate of tree loss in different zoning types in the next few decades because the tree protection ordinance applies to all new developments and redevelopments. Tree ordinances that are curated to their target region tend to protect and retroactively improve tree canopy coverage, as shown by several studies of the impact on ordinances on tree canopies in metropolitan areas in the United States (Hilbert et al. 2019; Pike et al. 2021; Hill, Dorfman, and Kramer 2010). For Greenville, there are several specific provisions related to parking lots that mandate minimum tree coverage requirements for parking lots, require raised parking median islands with trees and plants, and permit exemptions from minimum parking space regulation to protect heritage trees. Though not currently mandated, bioswales and rain gardens are encouraged by the ordinance and may eventually become mandated in future ordinances. In the final stages of drafting the ordinance, changes were made to increase tree and bush density requirements in commercial and multifamily residential areas, which will also help ensure that buffers between new developments include trees and other vegetation. Future work could include tracking the change in tree coverage throughout the city over time and understanding which zones receive the most gains and losses.

4.2. Proximity to street

Trees were most common immediately next to roads and least common at least 12 meters from roads. This is unsurprising because tree plantings near roads are often prioritized over planting further from roads. Trees tend to be planted near roadside sidewalks but far enough away that they do not interfere with power lines.

Space further from roads may lack trees due to a variety of reasons. In some areas, buildings are very close to roads, so trees cannot exist there. In other areas – usually the more suburban ones – development is often characterized by large buildings set back from roads with large parking lots between them and the roads. These parking lots may have some trees, but the space is dominated by impervious surfaces. In residential areas, trees are more likely to exist in all areas, but the location of houses can affect where trees are planted, and many homeowners select to have an open lawn in front of their homes.

The relatively lower percent tree coverage closer to streets – where sidewalks are usually located – is concerning because trees can have a notable impact on the microclimate of sidewalk spaces. In addition to the ecosystem services provided by trees that can benefit broader areas, trees have also been found to have notable improvements to air temperature, relative humidity, wind speed, and solar radiation exposure directly under the individual trees (Sanusi et al. 2017). The lower percent tree coverage closer to streets indicates that more needs to be done to plant and protect trees that are closer to the streets. Adding trees closer to streets will also have a greater impact on reducing the urban heat island effect by shading more paved surfaces.

4.3. Accuracy

By using manual tree verification of randomly sampled areas in Greenville, it was determined that the detection process was 53 percent accurate. The algorithm tended to overcount rather than undercount, with 71 percent of errors being type I errors, which occur when a tree location was incorrectly detected (false positive). The remaining 29 percent of errors were type II errors, which occur when a tree was missed by the detection process (false negative).

The 33 percent incorrect detection (false positive) rate is largely a symptom of the abundance of large deciduous trees in Greenville. Large deciduous trees have unpredictable, non-uniform shapes that can cause individual large branches to be misinterpreted as distinct trees; many of the larger deciduous trees were overcounted by the algorithm. False positives were also found along some fences and hedges, along the edges of sheds and detached garages, and on the edges of newly built decks.

This problem is complicated by the time and resolution of the LiDAR data provided by Greenville County. The LiDAR data was collected during the winter, which means deciduous trees did not have leaves when data were collected. This is typically beneficial for the County because it makes it easier to see buildings and roads, but it causes more gaps in the detected urban tree canopy. Because leaves are not present, fewer returns occur on the trees because the laser pulse travels past the tree. Additionally, the resolution of the LiDAR data provided by Greenville County is not nearly high enough. For Greenville's 2020 LiDAR data, the point density is 0.013 points per square meter – far lower than what is recommended in much of the scientific literature (Yun et al. 2021; Tanhuanpää et al. 2019; Popescu and Wynne 2004).

The usage of NDVI to determine the location of vegetation helped remove roads and parking lots and many remnants of buildings and walls that were not removed using building footprint data, but it was not always successful in removing everything that was not vegetation. NDVI is expressed as a raster of limited resolution, which results in some data being lost, whereas a point cloud retains high precision of point locations. Small, tall objects, such as telephone poles, street lights and fences, sometimes were not removed from the LiDAR data since they were not wide enough to be distinguished by the NDVI. This was not usually a problem with larger objects, such as parking lots, walls, and buildings.

Example causes of both main types of errors – errors due to deciduous tree shapes and errors due to unexpected ground surface objects – were revealed during the verification and accuracy determination process. Common examples of the overcounting of large deciduous trees can be seen in Figure 14, where the broad shape and branching pattern of deciduous trees or the non-uniform trimming pattern of deciduous trees can result in a single tree appearing to be multiple trees from an aerial angle.

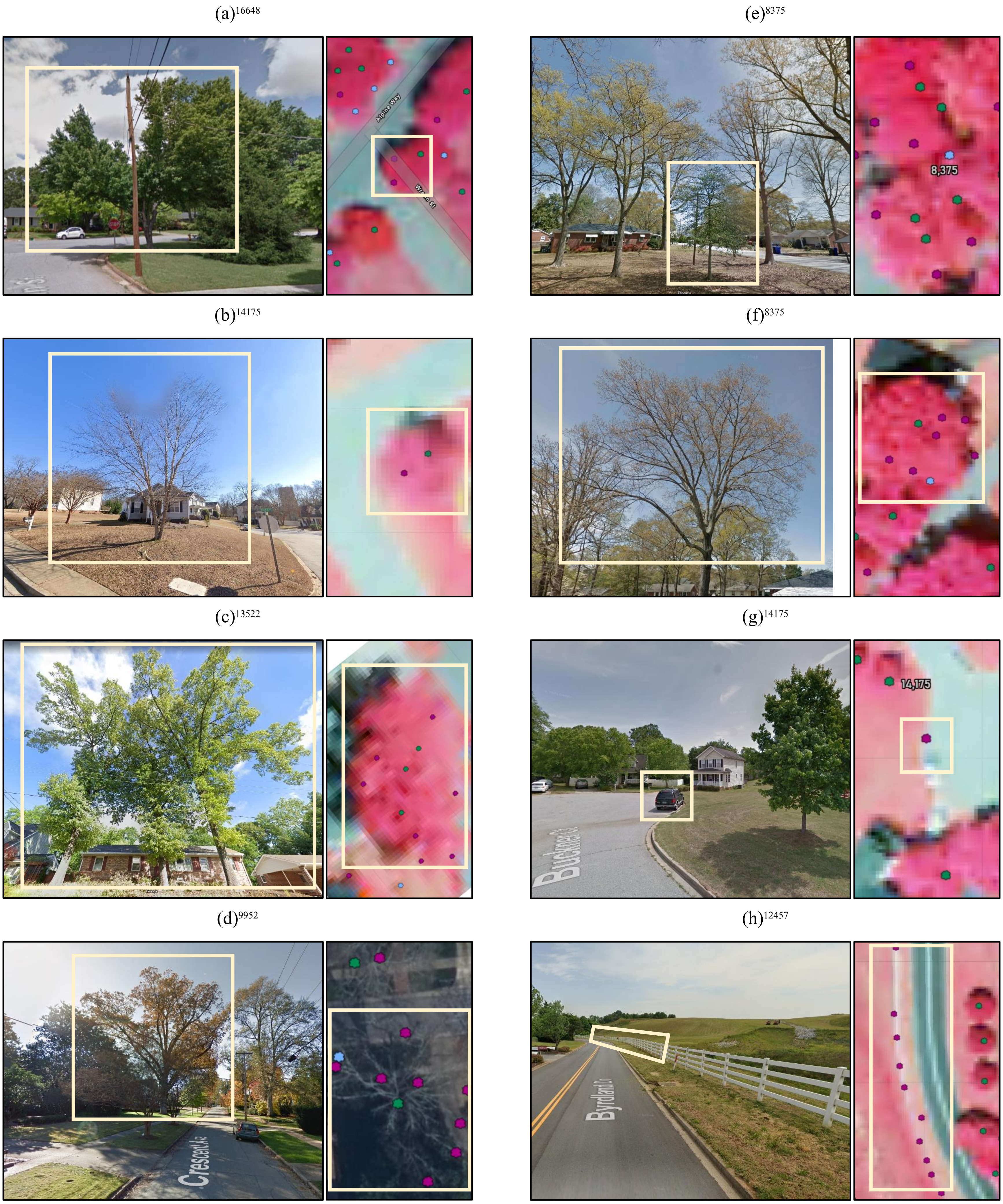


Figure 14. Selected examples of detection errors from the randomly selected verification area. Subfigures (a), (b), (c), (d), and (f) indicate false positives due to deciduous tree structure. Subfigure (e) represents a false negative due to a smaller tree being obscured by larger trees. Subfigure (g) exemplifies false positives due to some cars being incorrectly detected as trees. Subfigure (h) exemplifies false positives along fences when planar points are not removed during the LiDAR point cloud cleaning process. Superscript numbers in subfigure labels indicate the 100-meter fishnet grid from which the subfigure was selected. The symbology of the aerial images follows the symbology of Figure 7.

Most commonly, very large deciduous trees had visually distinguishable sections in their crowns where there were multiple clear local maxima for the entire tree crown. Since the algorithm works by identifying local maxima that are certain radius away from other local maxima, this is problematic with large deciduous trees that span a large horizontal distance compared to small- and medium-size trees.

Figure 14d shows a clear example of a single deciduous tree that has multiple clear crown sections that are each recognized as a separate tree by the detection algorithm. In a scenario where several large trees are more densely co-located, a large deciduous tree would have a much smaller crown radius and nearby separate local maxima would more likely represent a separate tree. However, where large deciduous are more isolated, the algorithm is too sensitive in assuming that nearby local maxima are separate trees.

When maintenance is performed on power lines, deciduous trees may also be cut so that branches are not in contact with the power lines. As a result, the tree can develop an unexpected shape, as shown in Figure 14a.

The remaining 14 percent missed tree (false negative) rate may occur due to obstruction of smaller trees or errors in the LiDAR point cloud filtering process. In urban settings, where large trees are routinely replaced by smaller trees when they outgrow their dedicated space, large trees can often mostly or completely cover smaller trees. When smaller trees are mostly obscured by larger trees, the detection process fails to detect them since the detection process evaluates the highest elevation points of the tree canopy.

For the detection process, an NDVI threshold was set that may have also excluded some smaller trees, which could also explain why some trees were incorrectly not detected. The threshold was a tradeoff between losing some vegetation or incorrectly keeping some paved surfaces and buildings.

Additionally, some trees were not detected because their points in the point cloud were erroneously removed. While the building footprints provided by Greenville County are generally accurate, their precision is not exact. As a result, during the LiDAR point cloud filtering process, a small buffer had to be created around the buildings, and all points within that buffer were removed from point cloud – even if the points represented a tree instead of a building. Conversely, sometimes the buffer extension was not enough, and points in the point cloud that represent buildings sometimes persisted through the detection process. Most buildings do not have trees immediately adjacent to their sides, so points from the buildings are often the tallest points, which results in the algorithm considering it to be the top of a tree.

The precision of detected tree points was not exact either. The detection process assumes that the highest point in a tree's crown indicates the center of the tree. However, the growth behavior of deciduous trees often results in the highest point of the tree being skewed in one direction, especially as the tree becomes larger. Visualizations of the detected trees make it abundantly clear that this is an issue, but it likely cannot be solved by an airborne remote sensing approach, which is why the analysis of the detected tree locations is focused on accuracy without consideration for precision.

To improve accuracy, an airborne remote sensing approach can potentially be paired with ground-level imagery to improve the precision of the process while also providing additional data about the trees. The usage of Google Street View has been evaluated by (Rodríguez-Puerta et al. 2022) and yields promising results in its ability to improve the placement of detected tree points and measure tree characteristics such DBH and tree species based on positional data and ground-based imagery. Google provides an API for retrieving Street View images for specific angles and times, and the temporal and spatial data associated with each Street View image could be used to identify the exact position of the trunk of each detected tree. Since the distance from the street would be accurately known, DBH could be deduced for each visible trunk. For trees that are sufficiently close enough to the street, classification algorithms could be applied to the leaves on each tree to automatically determine the species of each tree.

An AI-powered canopy density classification approach could also make it possible to optimize the detection radius for treetops for different growth densities. The "one size fits all" approach of applying a single detection process to a large area of diverse tree sizes and densities is unable to perform well in vastly different scenarios, but applying different algorithms to tree groups based on classified tree growth density may yield significantly better results when paired with higher-resolution LiDAR data. This method has been proposed and partially implemented in the past (Tanhuanpää et al. 2019) for other regions of the world, but it would also be more difficult to implement. In particular, Tanhuanpää et al. demonstrated that a small number of classified regions is not sufficient; instead, a classification process would need to be developed that has a sufficiently high number of classes to ensure that enough variations in tree structure are properly stratified (2019).

Recent advances in AI image recognition may also make it possible to more accurately identify trees regardless of tree size and density. While clipping and filtering of LiDAR point cloud data is still required to limit the point cloud to the extent of the forest, López Serrano et al. have demonstrated that AI-powered tree detection software could provide more specific and accurate tree detection results (2022). They claim a 97 percent accuracy rate for tree plots that contain varying tree density and size, but they do not specify the density of the LiDAR point clouds that were used in their testing. The software manual for the software used by López Serrano et al. indicates that higher resolution input data results in fewer detection errors (Dielmo 3D 2022). It is difficult to directly compare their accuracy rate with results of the project since the resolution of the LiDAR point cloud for their accuracy assessment is unknown and they evaluated accuracy in a rural area, but the ability of the AI-powered software to detect trees without additional segmentation of the input data based on tree density and size is very promising.

4.4. Census trends

Despite looking at a variety of census demographics factors – household income, education level, household size, median householder age, and unemployment – no clear trend emerged between any of these factors and vegetation coverage. Spatially, there appears to be trends between household income and vegetation coverage and household size and vegetation coverage for parts of the city, but the trend does not extend to the entire city. However, the lack of a city-wide trend at the census block level does not mean that there are not important data or trends to be found in the results.

Because of the lack of the city-wide trend, looking at Greenville's special emphasis neighborhoods is more helpful in this context: nearly all special emphasis neighborhoods lie within areas with lower income and relatively lower vegetation coverage (Figure 15). Since these neighborhoods are lower-income neighborhoods, the impact of not having trees has a more significant impact. For example, energy costs have a greater burden on lower-income households due to the fixed cost rate per kilowatt-hour of electricity, which causes energy costs to take a larger proportion of the household's month income (Kontokosta, Reina, and Bonczak 2020). Because of the fixed rate, the heat-reducing effect of tree shade has an outsized impact on lower-income households.

While adding trees adds many positive side-effects, it is important to recognize the gentrification potential when adding trees to neighborhoods. Many studies (Donovan et al. 2021; Rigolon and Németh 2020; Li 2019) have found that the addition of trees along streets, in yards, and in parks can gentrify neighborhoods by making them more desirable on the housing market, increasing property values to the point that only higher-income individuals or families can afford to own houses in the neighborhood. Additionally, it is important to recognize the costs of maintaining trees can often be challenging for lower-income homeowners. Support for tree maintenance could help homeowners be more interested in having trees on their property. The database of specific tree locations and heights generated by the individual tree detection process of this project will be valuable in identifying larger trees in lower-income areas that may be good targets for replacement by groups that advocate for tree planting and promotion, including TreesUpstate.

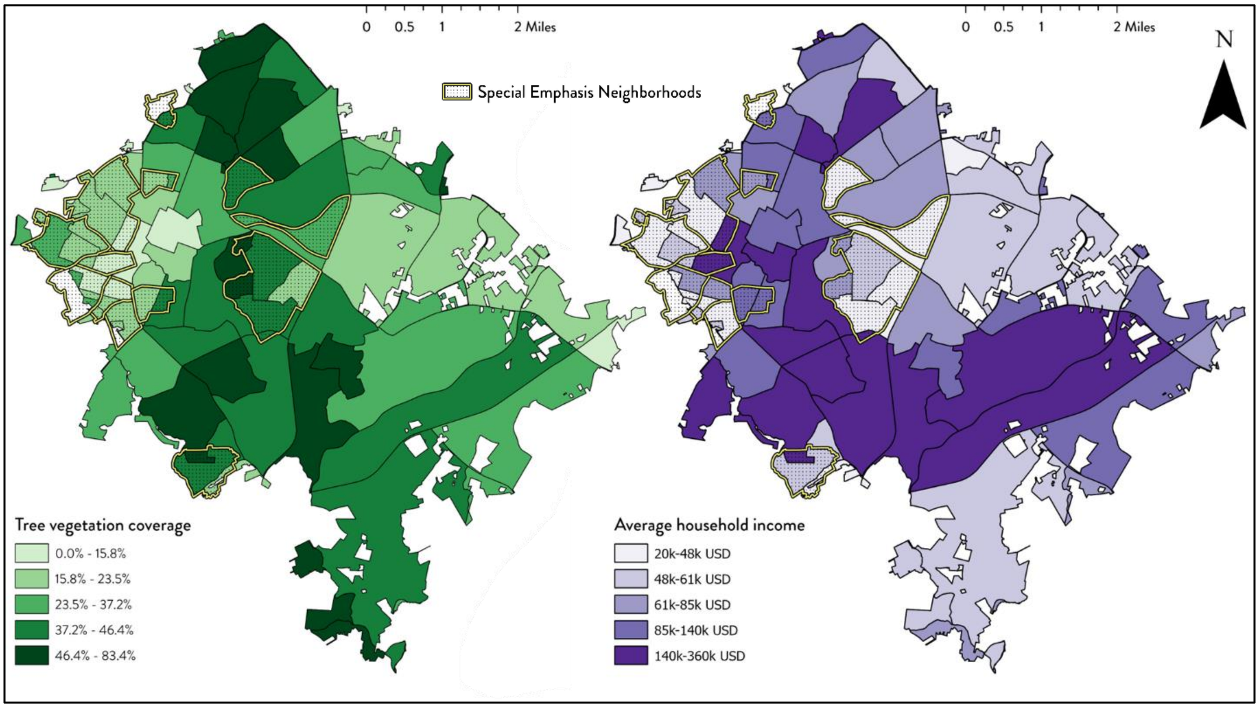


Figure 15. Tree vegetation coverage and average household income compared with reference to Greenville's special emphasis neighborhoods.

5. Conclusion

Automated tree detection in the City of Greenville using LiDAR yielded good results for smaller, isolated trees, but it struggled to accurately detect the locations of larger trees. The problems with detecting larger trees stem from the fact that the LiDAR data were collected in the winter, which meant that deciduous trees had no leaves, and the point cloud density was unsatisfactorily too low. Automated tree detection using LiDAR and GIS techniques will likely perform better with summertime LiDAR collected at higher resolution, and this detection process will likely perform well in other cities in the Southeastern United States region with high-density, summertime LiDAR.

Despite the need for improved accuracy of the detection process, the detected trees still provide a crucial initial database that gives a rough idea of tree coverage distribution throughout the detection region. Evaluated census trends indicated that higher-income single-family residential areas had the most coverage of trees, but the existence of the detected trees results create the opportunity for many different census-based analyses of tree distribution equity and other social-geographical tree-related projects.

Petitioning Greenville County or the City of Greenville to add higher-resolution summertime LiDAR collection to their remote sensing data collection initiatives is an obvious next step that can result in improved tree detection and tree coverage data. Recent efforts to use Google Street View in other locations to enhance the data obtained on collect streets (Rodríguez-Puerta et al. 2022) may also be helpful in improving the detection of trees in Greenville. Combined with improved results from higher-resolution LiDAR, evaluation and comparison of tree count and coverage at a smaller neighborhood scale may yield more-distinguishable differences in tree coverage and density between different neighborhoods that can spur more discussion and action to improve tree equity at the community level.

6. Acknowledgements

We would like to thank Patrick Flagler for providing advice on the use of ForestTools; Taylor Kite and Richard Hanning from the Greenville County GIS department for providing spatial data; and Dr. Geoffrey Habron, Dr. Matthew Cohen, and Dr. Ruth Aronoff for their encouragement, advice, and help during this project. We would also like to thank Jean-Romain Roussel for providing technical assistance with his LidR software package, Kylie Gambrill for data collection assistance, and Anna Justice and Drew Smith for sharing Greenville's TreePlotter data.

Thank you to Kylie Gambrill, Julia McCabe, Elizabeth Santi, Josh Kemp, Thomas McCoy-Bruce, and Savannah Steet for providing feedback during the research and writing process. Thank you to Kent Watkins for helping inspire this research.

Funding for this research was provided by Furman University through the Office of Undergraduate Research.

7. Appendices

7.1. Detected trees web map

The detected trees web map with up-to-date accuracy information and calculated fields is available at as an ArcGIS Instant App on the Furman University ArcGIS Online portal: https://furman.maps.arcgis.com/apps/instant/basic/index.html?appid=570201e58892483cb26c4749f1687048

7.2. Tree detection code

The following code was developed to detect trees from input false color composite imagery using the NIR, red, and green bands; building footprints; and LiDAR point clouds. It generates vegetation polygons, tree points, and rough tree crown polygons.

The code uses the `ɢõæɔẃ` (Esri 2022b) python package to integrate with ArcGIS Pro. The `ŞõðđŞņņƙr₈ɔẃŝ` file can be imported into ArcGIS Pro as a Python Toolbox.

The R (R Core Team 2022) code uses the `ɢõæəħrãħƪçħƪə` (Esri 2022a) package to integrate with ArcGIS Pro, the `ƙħçP` (Roussel et al. 2020; Roussel and Auty 2022) package to process LiDAR point clouds, the `õəçɢƙ` (Bivand, Keitt, and Rowlingson 2022) and `rẽ` (Pebesma 2018) packages to read and work with shapefiles, and the `çɔƙẃõ` (Wickham et al. 2022) package to manipulate data tables. The `PŞņņƙr₈ŝãw` file can be added as a R Toolbox.

The code follows the following folder-file structure:

```
treetools
|
├───.vscode
|       settings.json
|
├───python
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|   |   __init__.py
|   |
|   └───veg_poly
|           buffer_veg_poly.py
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|           create_veg_poly.py
|           create_veg_raster.py
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|           veg_raster_group.py
|
```

```
│           veg_raster_to_poly.py
│           __init__.py
│
└───R
        LidarTreeCrownDelineation.R
        PruneBuildingsFromLidar.R
        RTools.tbx
```

The contents of each file can be found in the following subsections. `__init__.py` should be an empty file; it is required by python to allow the `.py` files to be imported by other `.py` files.

Updated versions of the tree detection code are publicly available on GitHub in the jackbuehner/treetools repository.

7.2.1. python/DetectTrees.py

```python
import os
import shutil
from typing import List
import arcpy
import veg_poly

class DetectTrees(object):
  def __init__(self):
    self.label = 'Detect Trees from LAS'
    self.description = ''
    self.canRunInBackground = False

  #`
  # Define tool parameters
  # `
  def getParameterInfo(self):

    paramOrthoRaster = arcpy.Parameter(
       displayName='Orthophotography',
       name='ORTHO_RASTER',
       datatype='GPRasterLayer',
       parameterType='Required',
       direction='Input',
     )

    paramVegPoly = arcpy.Parameter(
       displayName='Vegetation Polygon',
       name='VEG_POLY',
       datatype='DEShapefile',
```

```
    parameterType='Optional',
    direction='Output',
    category='Intermediate Outputs',
  )

  paramVegPolyBuffered = arcpy.Parameter(
    displayName='Buffered Vegetation Polygon',
    name='VEG_POLY_BUFFERED',
    datatype='DEShapefile',
    parameterType='Optional',
    direction='Output',
    category='Intermediate Outputs',
  )

  paramBuildings = arcpy.Parameter(
    displayName='Buildings',
    name='BUILDINGS',
    datatype='DEShapefile',
    parameterType='Required',
    direction='Input'
  )

  paramBuildingsBuffered = arcpy.Parameter(
    displayName='Buffered Buildings',
    name='BUILDINGS_BUFFERED',
    datatype='DEShapefile',
    parameterType='Optional',
    direction='Output',
    category='Intermediate Outputs'
  )

  paramLAS = arcpy.Parameter(
    displayName='LAS Dataset',
    name='LAS_DATASET',
    datatype='DELasDataset',
    parameterType='Required',
    direction='Input'
  )

  paramFilteredLasDataset = arcpy.Parameter(
    displayName='Filtered LAS dataset',
    name='FILTERED_LAS_DATASET',
    datatype='DELasDataset',
    parameterType='Optional',
    direction='Output',
```

```
    category='Intermediate Outputs',
  )

paramDsm = arcpy.Parameter(
    displayName='Digital Surface Model',
    name='DSM',
    datatype='GPRasterLayer',
    parameterType='Optional',
    direction='Output',
    category='Intermediate Outputs',
  )

paramTreePoints = arcpy.Parameter(
    displayName='Tree Points',
    name='TREE_POINTS',
    datatype='DEShapefile',
    parameterType='Required',
    direction='Output',
  )

paramTreeCrownPolygons = arcpy.Parameter(
    displayName='Tree Crown Polygons',
    name='TREE_CROWN_POLYGONS',
    datatype='DEShapefile',
    parameterType='Optional',
    direction='Output',
    category='Intermediate Outputs',
  )

paramExtent = arcpy.Parameter(
    displayName='Processing extent shapefile',
    name='EXTENT_SHAPE',
    datatype='DEShapefile',
    parameterType='Required',
    direction='Input'
  )

paramDsmCellSize = arcpy.Parameter(
    displayName='DSM Cell Size',
    name='DSM_CELLSIZE',
    datatype='Long',
    parameterType='Required',
    direction='Input',
  )
paramDsmCellSize.value = 2
```

```
paramMinTreeHeight = arcpy.Parameter(
   displayName='Minimum Tree Height',
   name='MIN_TREE_HEIGHT',
   datatype='Long',
   parameterType='Required',
   direction='Input',
  )
paramMinTreeHeight.value = 2

paramMinTreeCrownHeight = arcpy.Parameter(
   displayName='Minimum Tree Crown Height',
   name='MIN_TREE_CROWN_HEIGHT',
   datatype='Long',
   parameterType='Required',
   direction='Input',
  )
paramMinTreeCrownHeight.value = 2

paramWindowFunction = arcpy.Parameter(
   displayName='Window Function',
   name='WINDOW_FUNCTION',
   datatype='GPString',
   parameterType='Required',
   direction='Input',
  )
paramWindowFunction.value = '0.00901*x + 2.51513'

paramBufferSize = arcpy.Parameter(
   displayName='Buffer Size',
   name='BUFFER_SIZE',
   datatype='GPLinearUnit',
   parameterType='Required',
   direction='Input',
  )
paramBufferSize.value = '1 meter'

parameters = [
  paramOrthoRaster,
  paramLAS,
  paramBuildings,
  paramExtent,
  paramBufferSize,
  paramDsmCellSize,
  paramMinTreeHeight,
```

```
      paramMinTreeCrownHeight,
      paramWindowFunction,
      paramTreePoints,
      paramDsm,
      paramVegPoly,
      paramVegPolyBuffered,
      paramBuildingsBuffered,
      paramFilteredLasDataset,
      paramTreeCrownPolygons,
    ]

    return parameters

  #`
  # Whether this tool is licensed to run. We check the required
  # ArcGIS Pro licenses.
  # `
  def isLicensed(self):
    try:
      if arcpy.CheckExtension('Spatial') != 'Available': raise Exception
    except Exception: return False # The tool cannot be run
    return True # The tool can be run

  #`
  # Modify the values and properties of parameters before internal
  # validation is performed. This method is called whenever a parameter
  # has been changed.
  # `
  def updateParameters(self, parameters: List[arcpy.Parameter]):
    return

  #`
  # Modify the messages created by internal validation for each tool
  # parameter. This method is called after internal validation.
  # `
  def updateMessages(self, parameters):
    return

  def execute(self, parameters: List[arcpy.Parameter], messages):
    # import the custom toolbox containing the R scripts
    arcpy.ImportToolbox(f'{arcpy.env.packageWorkspace}/../R/RTools', 'rtools')

    # create a dictionary containing key-value pairs of the parameters
    # where the key is the param name and the value is the text value
    params = {}
```

```
    for elem in parameters:
      if (elem.altered):
        params[elem.name] = elem.valueAsText

    # create temporary folders
    TEMP_PATH = f'{arcpy.env.scratchFolder}/DetectTrees'
    if os.path.exists(TEMP_PATH): shutil.rmtree(TEMP_PATH)
    os.mkdir(TEMP_PATH)
    VEG_FOLDER = f'{TEMP_PATH}/veg'
    TILED_LAS_FOLDER = f'{TEMP_PATH}/tiled_las'
    CLIPPED_LAS_FOLDER = f'{TEMP_PATH}/clipped_las'
    FILTERED_LAS_FOLDER = f'{TEMP_PATH}/filtered_las'
    CLIPPED_BUILDINGS_FOLDER = f'{TEMP_PATH}/clipped_buildings'
    os.mkdir(VEG_FOLDER)
    os.mkdir(TILED_LAS_FOLDER)
    os.mkdir(CLIPPED_LAS_FOLDER)
    os.mkdir(FILTERED_LAS_FOLDER)
    os.mkdir(CLIPPED_BUILDINGS_FOLDER)

    # create vegetation polygons based on orthophotography with NIR and red bands
    # so we can target the areas with vegetation
    ORTHO_RASTER = params.get('ORTHO_RASTER')
    VEG_POLY = params.get('VEG_POLY', f'{VEG_FOLDER}/veg_poly')
    VEG_POLY_BUFFERED = params.get('VEG_POLY_BUFFERED',
f'{VEG_FOLDER}/veg_poly_buffered')
    EXTENT_SHAPE = params.get('EXTENT_SHAPE')
    CLIPPED_ORTHO_RASTER = params.get('CLIPPED_ORTHO_RASTER',
f'{TEMP_PATH}/ortho.tif')
    BUFFER_SIZE = params.get('BUFFER_SIZE')
    arcpy.AddMessage('Clipping orthophoto raster to extent...')
    arcpy.management.Clip(ORTHO_RASTER, None, CLIPPED_ORTHO_RASTER, EXTENT_SHAPE,
'256', 'ClippingGeometry', 'MAINTAIN_EXTENT') # clip to input extent to reduce
processing time
    arcpy.AddMessage('Generating vegetation raster...')
    veg_poly.create(CLIPPED_ORTHO_RASTER, VEG_POLY, VEG_POLY_BUFFERED,
BUFFER_SIZE)

    # clip the input LAS dataset to the provided extent
    # and convert it to non-overlapping square kilometer LAS tiles
    LAS_DATASET = params.get('LAS_DATASET')
    LAS_TILED_DATSET = f'{TILED_LAS_FOLDER}/tiles.lasd'
    LAS_CLIPPED_DATSET = f'{TEMP_PATH}/clipped_las/tiles.lasd'
    arcpy.AddMessage('Clapping LAS dataset to extent...')
```

```
    arcpy.ddd.ExtractLas(LAS_DATASET, CLIPPED_LAS_FOLDER, 'MAXOF', EXTENT_SHAPE,
'PROCESS_EXTENT', '', 'MAINTAIN_VLR', 'REARRANGE_POINTS', 'NO_COMPUTE_STATS',
LAS_CLIPPED_DATSET, 'NO_COMPRESSION')
    arcpy.AddMessage('Tiling clipped LAS dataset...')
    arcpy.ddd.TileLas(LAS_CLIPPED_DATSET, TILED_LAS_FOLDER, 'tile',
LAS_TILED_DATSET, 'COMPUTE_STATS', '1.4', None, 'NO_COMPRESSION',
'REARRANGE_POINTS', None, 'ROW_COLUMN', None, '1 kilometer', '1 kilometer', None)

    # loop through the clipped las tiles to generate a list of las files to
process
    las_file_names = []
    for fileName in os.listdir(TILED_LAS_FOLDER):
      if (fileName[-4:len(fileName)] == '.las'): las_file_names.append(fileName)

    # create lidar without noise from buildings, non-vegetated areas, and planar
surfaces
    BUILDINGS = params.get('BUILDINGS')
    BUILDINGS_CRS = arcpy.Describe(BUILDINGS).spatialReference
    BUILDING_ELEV_ATTR = 'TOPELEV'
    BUILDING_ELEV_ADD = 5
    FILTERED_LAS_DATASET = params.get('FILTERED_LAS_DATASET',
f'{FILTERED_LAS_FOLDER}/tiles.lasd')

    for index, las in enumerate(las_file_names):
      arcpy.AddMessage(f'Filtering {las} ({index + 1}/{len(las_file_names)})...')
      input_las = f'{TILED_LAS_FOLDER}/{las}'

      # create polygon boundary for each tile
      las_tile_boundary = f'{TILED_LAS_FOLDER}/{las[0:-4]}.shp'
      arcpy.AddMessage('...generating tile boundary')
      arcpy.ddd.PointFileInformation(input_las, las_tile_boundary, 'LAS', '',
BUILDINGS_CRS, 'NO_RECURSION')

      # clip buildings to las tile and then buffer them
      buildings_clipped = f'{CLIPPED_BUILDINGS_FOLDER}/{las[0:-4]}.shp'
      arcpy.AddMessage('...clipping buildings to tile')
      arcpy.analysis.PairwiseClip(BUILDINGS, las_tile_boundary,
buildings_clipped);
      buildings_buffered = f'{CLIPPED_BUILDINGS_FOLDER}/{las[0:-4]}_buffered.shp'
      arcpy.AddMessage('...buffering buildings')
      arcpy.Buffer_analysis(buildings_clipped, buildings_buffered, BUFFER_SIZE,
'FULL', 'ROUND', 'NONE')

      # clean las tile
      output_las = f'{FILTERED_LAS_FOLDER}/{las}'
```

```
     arcpy.AddMessage('...cleaning las tile')
     arcpy.rtools.PruneBuildingsFromLidar(input_las, buildings_buffered,
output_las, BUILDING_ELEV_ATTR, BUILDING_ELEV_ADD, True, VEG_POLY_BUFFERED,
True)

   output_las_files = []
   for fileName in os.listdir(FILTERED_LAS_FOLDER):
output_las_files.append(f'{FILTERED_LAS_FOLDER}/{fileName}')
   arcpy.AddMessage('Creating filtered LAS dataset from filtered tiles...')
   arcpy.management.CreateLasDataset(output_las_files, FILTERED_LAS_DATASET)

   # convert lidar points to a digital surface model (DSM)
   DSM = f'{TEMP_PATH}/dsm.tif'
   DSM_CLIPPED = params.get('DSM', f'{TEMP_PATH}/dsm_clipped.tif')
   DSM_CELLSIZE = params.get('DSM_CELLSIZE')
   arcpy.AddMessage('Converting filtered LAS dataset to digital surface model
(DSM)...')
   arcpy.conversion.LasDatasetToRaster(FILTERED_LAS_DATASET, DSM, 'ELEVATION',
'BINNING MAXIMUM NATURAL_NEIGHBOR', 'FLOAT', 'CELLSIZE', DSM_CELLSIZE)
   arcpy.AddMessage('Clipping DSM to extent...')
   arcpy.management.Clip(DSM, None, DSM_CLIPPED, EXTENT_SHAPE, '256',
'ClippingGeometry', 'NO_MAINTAIN_EXTENT'); # clip to extent to save processing
time

   # generate tree points and tree crowns
   # (tree crowns are skipped if no output path is provided)
   TREE_POINTS = params.get('TREE_POINTS')
   TREE_CROWN_POLYGONS = params.get('TREE_CROWN_POLYGONS', None)
   MIN_TREE_HEIGHT = params.get('MIN_TREE_HEIGHT')
   MIN_TREE_CROWN_HEIGHT = params.get('MIN_TREE_CROWN_HEIGHT')
   WINDOW_FUNCTION = params.get('WINDOW_FUNCTION')
   arcpy.AddMessage('Detecting trees...')
   arcpy.rtools.LidarTreeCrownDelineation(DSM_CLIPPED, MIN_TREE_HEIGHT,
WINDOW_FUNCTION, TREE_POINTS, MIN_TREE_CROWN_HEIGHT, TREE_CROWN_POLYGONS)

   return

  #`
  # This method takes place after outputs are outputs are processed and
  # added to the display.
  # `
  def postExecute(self, parameters):
   return
```

7.2.2. python/TreeTools.pyt

```
import importlib

# this convoluted way of importing the tools is required
# because ArcGIS does not reload imported modules
# unless the software is restarted
import DetectTrees
importlib.reload(DetectTrees)
from DetectTrees import DetectTrees as DetectTreesTool

class Toolbox(object):
  def __init__(self):
    self.label = "Tree Tools"
    self.alias = "treetools"

    # List of tool classes associated with this toolbox
    self.tools = [DetectTreesTool]
```

7.2.3. veg_poly/buffer_veg_poly.py

```
import sys
import arcpy

def buffer_veg_poly(veg_poly_shp_path: str, out_path: str, size: str) -> str:
    sys.stdout.write('Generating vegetation polygon buffer...')
    arcpy.Buffer_analysis(veg_poly_shp_path, out_path, size, "FULL", "ROUND",
"NONE")
    sys.stdout.write(' ✓ Done\n')
    return out_path
```

7.2.4. veg_poly/create_ndvi.py

```
import sys
import arcpy

def create_ndvi(ortho: str, nir_band: int, red_band: int) -> arcpy.Raster:
  sys.stdout.write('Creating normalized difference vegetation index (NDVI)...')
  ndvi = arcpy.sa.BandArithmetic(ortho, f'{nir_band} {red_band}', 1)
  sys.stdout.write(' ✓ Done\n')
  return ndvi
```

7.2.5. veg_poly/create_veg_poly.py

```
from typing import Tuple
import arcpy
from .create_ndvi import create_ndvi;
```

```
from .create_veg_raster import create_veg_raster;
from .veg_raster_group import veg_raster_group;
from .veg_raster_cleanup import veg_raster_cleanup;
from .veg_raster_to_poly import veg_raster_to_poly
from .buffer_veg_poly import buffer_veg_poly;

class SpatialLicenseError(Exception): pass

def create_veg_poly(ortho_raster_path: str, veg_poly_out_path: str,
veg_poly_buffer_out_path: str, veg_poly_buffer_size: str) -> Tuple[str, str]:
  try:
    # permit using the spatial analyist license from arcgis pro
    if arcpy.CheckExtension('Spatial') == 'Available':
arcpy.CheckOutExtension('Spatial')
    else: raise SpatialLicenseError

    print('\n⌛ Generating vegetation polygons')

    # create ndvi from orthophotography raster containing NIR and red bands
    ndvi = create_ndvi(ortho_raster_path, 1, 2)

    # create a vegetation raster
    veg_raster = create_veg_raster(ndvi)
    veg_raster_grouped = veg_raster_group(veg_raster)
    veg_raster_grouped = veg_raster_cleanup(veg_raster_grouped)

    # create shapefiles
    arcpy.env.overwriteOutput = True
    out = veg_raster_to_poly(veg_raster_grouped, veg_poly_out_path)
    buffer_out = buffer_veg_poly(veg_poly_out_path, veg_poly_buffer_out_path,
veg_poly_buffer_size)
    print('✅ DONE')
    return [out, buffer_out]

  except SpatialLicenseError: print('Spatial license in unavailable')
  except arcpy.ExecuteError: print(arcpy.GetMessages(2))

  finally:
    arcpy.CheckInExtension('Spatial')
```

7.2.6. veg_poly/create_veg_raster.py

```
import sys
import arcpy
```

```
def create_veg_raster(raster: arcpy.Raster) -> arcpy.Raster:
   sys.stdout.write('Creating initital vegetation raster...')
   raster = arcpy.sa.RasterCalculator([raster], ['x'], 'x > 0.1')
   sys.stdout.write(' ✓ Done\n')
   return raster
```

7.2.7. veg_poly/veg_raster_cleanup.py

```
import sys
import arcpy

def veg_raster_cleanup(raster: arcpy.Raster) -> arcpy.Raster:
   sys.stdout.write('Removing non-vegetation pixels and isolated vegetation
pixels...')
   new_raster = arcpy.sa.ExtractByAttributes(raster, 'Count > 3 And LINK = 1')
   sys.stdout.write(' ✓ Done \n')
   return new_raster
```

7.2.8. veg_poly/veg_raster_group.py

```
import sys
import arcpy

def veg_raster_group(raster: arcpy.Raster) -> arcpy.Raster:
   sys.stdout.write('Grouping matching vegetation raster pixels...')
   raster = arcpy.sa.RegionGroup(raster, 'FOUR', 'WITHIN', True, 0)
   sys.stdout.write(' ✓ Done\n')
   return raster
```

7.2.9. veg_poly/veg_raster_to_poly.py

```
import sys
import arcpy

def veg_raster_to_poly(raster: arcpy.Raster, out_path: str) -> str:
   sys.stdout.write('Converting filtered vegetation raster to polygon...')
   arcpy.RasterToPolygon_conversion(raster, out_path, False, 'Value', True)
   sys.stdout.write(' ✓ Done\n')
   return out_path
```

7.3. Extending zoning boundaries into the center of roads divididing zones

Greenville County and the City of Greenville collaborate on their zoning GIS data; the City of Greenville simply provides a clipped version of the Greenville County zoning data polygons. This zoning data, however, omits roads and area directly next to roads (Figure 16). In order to report the proportion of detected trees that fall into each zone, it is necessary to extend the zoning polygons to meet in the center of the roads.

The City and County provide polygons that represent the extent of the roads in the City of Greeenville, but the road and extent and the zoning polygons do not share borders. Therefore, in order to fill the voids between the zoning polygons, a union operation was performed between the City of Greenville boundary and the zoning polygons. The zoning attributes from the zoning polygons were retained, which meant that the space between each zone could be distinguished by empty zoning data.

For the generated polygons that represent the voids between zones, a centerline was drawn. This centerline follows the exact center of the voids between zones, which generally also follows the center of roads. This centerline was then used to split the polygons that represent the voids between zones.

To apply zoning data to the voids between zones, the Eliminate tool from ArcGIS Pro was used with the road void polygons and the zoning polygons. Each road void polygon was assigned the same attributes as the zoning polygon that had the most shared perimeter. All polygons with the same attributes were merged (dissolved) such that no polygons with the same attributes could exists next to each other. An example of the final output can be seen in Figure 17 and compared against Figure 16.

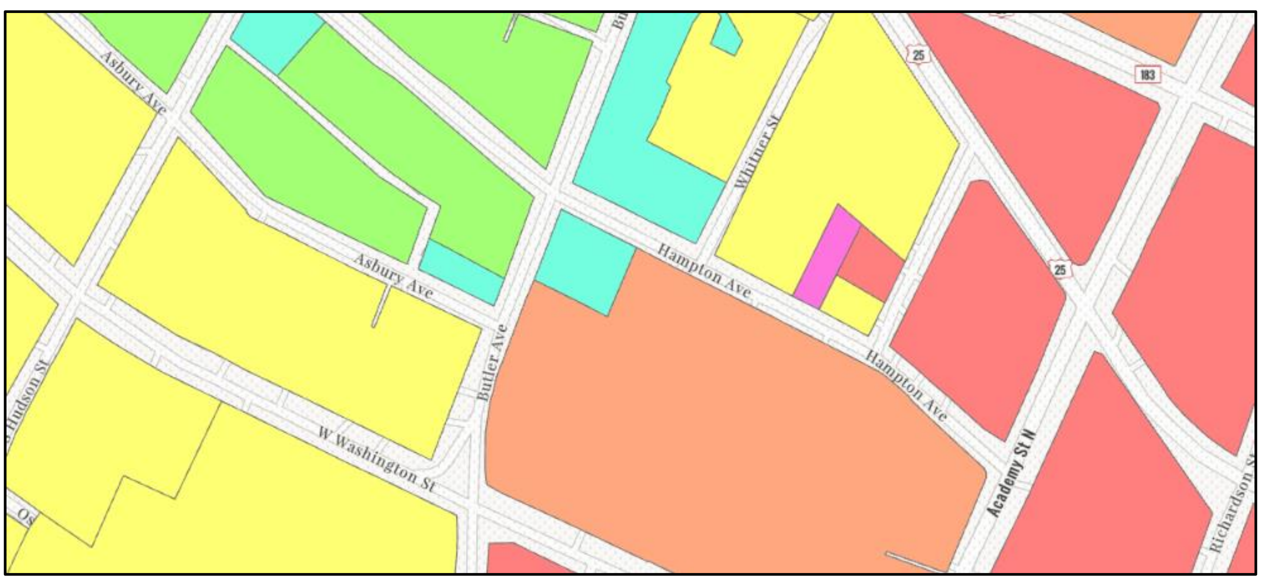

Figure 16. An example of provided zoning data from Greenville County and the City of Greenville. Zones do not quite extend to the edges of roads. Yellow represents planned development districts, orange and red represent commercial districts, turquoise represents residential districts, and pink represents office and institutional districts.

Figure 17. An example of the provided zoning data once the voids between zones are filled. This example has the same area and symbology as Figure 16. Voids are filled to the center of the roads with the zone with the most shared perimeter. In most cases, zones are joined at the center of roads. In some cases, zones may wrap around other zones. This can be seen in the center of the figure around the turquoise residential district.

7.4. Vegetation analysis for zones

The output from appendix section 7.3 was used to determine the percent vegetation coverage, the percent coverage of vegetation for each zone, and the share of vegetation across all zones. Certain zones, such as single-family residential districts, have multiple sub-types that indicate different minimum lot sizes and other parameters ("County of Greenville Zoning Ordinance" 2023). These zones were collapsed into zones of the same name for ease of analysis.

In ArcGIS Pro 3.0, an intersection between the zoning data with removed voids and polygons representing where vegetation is located was performed. This adds the zoning attributes to the vegetation polygon. The vegetation polygon is split into different polygons during this process so that the zoning attributes can be applied only where there is spatial overlap. This process also calculates the length and area of each polygon.

The attribute tables for the zoning layer and the resultant intersection layer were exported to comma-separate values (CSV) files. The following R (R Core Team 2022) code was created to process the data from the attribute tables and generate relevant figures. The `tidyverse` (Wickham et al. 2019) package was used to assist with data manipulation.

```
library(tidyverse)
```

```r
original_data = read_csv('data/veg poly intersect.csv')

# create a new tibble that only includes the columns
# we need and rename them to clearly describe their
# meanings
veg_data = original_data %>%
  transmute(
    area = SHAPE_Area,
    zone = as.factor(case_when(
      ZONING == 'R-6' ~ 'Single-family residential',
      ZONING == 'R-7.5' ~ 'Single-family residential',
      ZONING == 'R-9' ~ 'Single-family residential',
      ZONING == 'R-10' ~ 'Single-family residential',
      ZONING == 'R-12' ~ 'Single-family residential',
      ZONING == 'R-20' ~ 'Single-family residential',
      ZONING == 'R-M' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M1' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M1.5' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M2' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M3' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M7.5' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M10' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M12' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M20' ~ 'Single-family and multifamily residential',
      ZONING == 'R-MA' ~ 'Multifamily residential',
      ZONING == 'R-S' ~ 'Suburban residential',
      ZONING == 'OD' ~ 'Office and institutional district',
      ZONING == 'C-1' ~ 'Neighborhood commercial district',
      ZONING == 'C-2' ~ 'Commercial district',
      ZONING == 'C-3' ~ 'Commercial district',
      ZONING == 'C-4' ~ 'Commercial district',
      ZONING == 'S-1' ~ 'Service district',
      ZONING == 'I-1' ~ 'Industrial district',
      ZONING == 'O-D' ~ 'Office and institutional district',
      ZONING == 'PD' ~ 'Planned development district',
      ZONING == 'RDV' ~ 'Redevelopment district',
      ZONING == 'PO' ~ 'Preservation overlay district',
      ZONING == 'NRO' ~ 'Neighborhood revitalization overlay district',
      ZONING == 'RR-ROW' ~ 'Railroad',
      ZONING == 'SSOD' ~ 'Special Sign Overlay District',
      ZONING == 'AP' ~ 'Airport protective overlay district',
      ZONING == 'SFHA' ~ 'Special flood hazard area overlay district',
      ZONING == 'FRD' ~ 'Flexible review district',
      TRUE ~ 'NA',
    )),
```

```
  ) %>%
  filter(zone != 'NA')

total_veg_area = sum(data$area)

# read the zone shape data and get the area and zone name
zone_data = read_csv('data/GreenvilleCityZoningDissolved2020_ExportTable.csv')
%>%
  transmute(
    area = SHAPE_Area,
    zone = as.factor(case_when(
      ZONING == 'R-6' ~ 'Single-family residential',
      ZONING == 'R-7.5' ~ 'Single-family residential',
      ZONING == 'R-9' ~ 'Single-family residential',
      ZONING == 'R-10' ~ 'Single-family residential',
      ZONING == 'R-12' ~ 'Single-family residential',
      ZONING == 'R-20' ~ 'Single-family residential',
      ZONING == 'R-M' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M1' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M1.5' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M2' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M3' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M7.5' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M10' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M12' ~ 'Single-family and multifamily residential',
      ZONING == 'R-M20' ~ 'Single-family and multifamily residential',
      ZONING == 'R-MA' ~ 'Multifamily residential',
      ZONING == 'R-S' ~ 'Suburban residential',
      ZONING == 'OD' ~ 'Office and institutional district',
      ZONING == 'C-1' ~ 'Neighborhood commercial district',
      ZONING == 'C-2' ~ 'Commercial district',
      ZONING == 'C-3' ~ 'Commercial district',
      ZONING == 'C-4' ~ 'Commercial district',
      ZONING == 'S-1' ~ 'Service district',
      ZONING == 'I-1' ~ 'Industrial district',
      ZONING == 'O-D' ~ 'Office and institutional district',
      ZONING == 'PD' ~ 'Planned development district',
      ZONING == 'RDV' ~ 'Redevelopment district',
      ZONING == 'PO' ~ 'Preservation overlay district',
      ZONING == 'NRO' ~ 'Neighborhood revitalization overlay district',
      ZONING == 'RR-ROW' ~ 'Railroad',
      ZONING == 'SSOD' ~ 'Special Sign Overlay District',
      ZONING == 'AP' ~ 'Airport protective overlay district',
      ZONING == 'SFHA' ~ 'Special flood hazard area overlay district',
      ZONING == 'FRD' ~ 'Flexible review district',
```

```
    TRUE ~ 'NA',
   )),
  ) %>%
  filter(zone != 'NA') %>%
  group_by(zone) %>%
  summarise(
   area = sum(area),
  )

# determine vegetation area information for each zone
grouped_veg_data = veg_data %>%
  group_by(zone) %>%
  summarise(
   # total area of vegetation for the zone
   veg_area = sum(area),
   # percent of the total vegetation area for the city
   percent_share_veg_area = (sum(area) / total_area) * 100
  )

# join the tibbles so we have vegetation and zone area in one table
zone_veg_data = full_join(zone_data, grouped_veg_data, by = 'zone')

# calculate the percent of each zone that is covered in vegetation
zone_veg_data = zone_veg_data %>%
  mutate(
   percent_veg_area = (veg_area / area) * 100
  )

# determine percent vegetation coverage for the all zones
total_veg_area = sum(zone_veg_data$veg_area)
total_area = sum(zone_veg_data$area)
percent_veg_coverage = (total_veg_area / total_area) * 100
percent_veg_coverage


# plot the share of vegetation coverage for each zone
plot1a = zone_veg_data %>%
  ggplot(aes(y = zone, x = percent_share_veg_area)) +
  geom_bar(
   stat = 'identity', # use x aesthetic for value,
   col = 'black',
   fill = 'black'
  ) +
  theme_bw() +
  xlab('Percent share of vegetation coverage') +
```

```
  ylab('')

# plot the percent vegetation coverage for each zone
plot1b = zone_veg_data %>%
  ggplot(aes(y = zone, x = percent_veg_area)) +
  geom_bar(
    stat = 'identity', # use x aesthetic for value,
    col = 'black',
    fill = 'black'
  ) +
  theme_bw() +
  xlab('Percent vegetation coverage') +
  ylab('')
```

7.5. Vegetation coverage calculation for buffers

To calculate the vegetation area coverage for different distances from the roads, three different buffers were drawn with ArcGIS Pro: a 2-meter buffer from the roads, a 12-meter buffer from the roads, and a 25-meter buffer from the roads. Then, spatial data indicating the location of tree vegetation was used to tabulate the area of each buffer that represented vegetated and non-vegetated area. These areas, however, overlap. For example, the 12-meter buffer includes the 2-meter buffer.

The following R (R Core Team 2022) code was used to calculate the percentage tree vegetation coverage for each buffer with overlapping area removed. The tabulated vegetation area for each buffer was imported, and areas of subset buffers were subtracted. The data was then joined, percent coverage was calculated, and a bar chart was produced. The `ŝħçŵūðõrð` (Wickham et al. 2019) package was used to assist with data manipulation.

```
library(tidyverse)

# import road buffer vegetation,
# transmuting to include Buffer name and Percent coverage for
# vegetated and non-vegetated area
road_buffer_veg_data_2m = read_csv('data/RoadBuffer2mVegArea.csv') %>%
  transmute(
    Buffer = '2m',
    Vegetated = case_when(Value == 1 ~ TRUE, TRUE ~ FALSE),
    Area = Area,
    #Percent = (Area / (Area[[1]] + Area[[2]])) * 100,
    RangeArea = Area,
  )
road_buffer_veg_data_12m = read_csv('data/RoadBuffer12mVegArea.csv') %>%
```

```
  transmute(
    Buffer = '12m',
    Vegetated = case_when(Value == 1 ~ TRUE, TRUE ~ FALSE),
    Area = Area,
    # get the area, but subtract the area from the previous buffer that
    # is a subset of this buffer
    RangeArea = Area - filter(
                  road_buffer_veg_data_2m,
                  Vegetated == case_when(
                   Value == 1 ~ TRUE,
                   TRUE ~ FALSE)
                  )$Area
  )
road_buffer_veg_data_25m = read_csv('data/RoadBuffer25mVegArea.csv') %>%
  transmute(
    Buffer = '25m',
    Vegetated = case_when(Value == 1 ~ TRUE, TRUE ~ FALSE),
    Area = Area,
    # get the area, but subtract the area from the previous buffers that
    # are subsets of this buffer
    RangeArea = Area
              - filter(
                  road_buffer_veg_data_2m,
                  Vegetated == case_when(
                    Value == 1 ~ TRUE,
                    TRUE ~ FALSE
                  )
                )$Area
              - filter(
                  road_buffer_veg_data_12m,
                  Vegetated == case_when(
                    Value == 1 ~ TRUE,
                    TRUE ~ FALSE
                  )
                )$Area
  )

# join the buffers into a single tibble
road_buffer_veg_data = full_join(road_buffer_veg_data_25m,
road_buffer_veg_data_12m)
road_buffer_veg_data = full_join(road_buffer_veg_data, road_buffer_veg_data_2m)

# calculate the percent vegetation coverage for each buffer
# using the range area instead of the entire buffer area
road_buffer_veg_data = road_buffer_veg_data %>%
```

```
 mutate(Percent = (Area / (Area[[1]] + Area[[2]])) * 100)
road_buffer_veg_data

# coverage percent for different proximities
road_buffer_veg_data %>%
 mutate(Buffer = case_when(
  Buffer == '2m' ~ 'Along street [0, 2)',
  Buffer == '12m' ~ 'Close to street [2, 12)',
  Buffer == '25m' ~ 'Further from street [12, 25]',
  TRUE ~ '',
 )) %>%
 filter(Vegetated == TRUE) %>%
 ggplot(aes(y = Buffer, x = Percent)) +
  geom_bar(
   stat = 'identity', # use x aesthetic for value
   col = 'black',
   fill = 'lightblue'
  ) +
  theme_bw() +
  xlab('Percent') +
  ylab('Proximity')
```

8. References

"About I-Tree." n.d. I-Tree Tools. Accessed April 17, 2023. https://www.itreetools.org/about.

American Forests, and EarthDefine. 2020. "Tree Equity Score Report for Greenville, SC." Tree Equity Score. 2020. https://www.treeequityscore.org/reports/place/greenville-sc/.

Angelo, Hillary, Key MacFarlane, James Sirigotis, and Adam Millard-Ball. 2022. "Missing the Housing for the Trees: Equity in Urban Climate Planning." *Journal of Planning Education and Research*, January, 0739456X211072527. https://doi.org/10.1177/0739456X211072527.

Awrangjeb, Mohammad, Mehdi Ravanbakhsh, and Clive S. Fraser. 2010. "Automatic Detection of Residential Buildings Using LIDAR Data and Multispectral Imagery." *ISPRS Journal of Photogrammetry and Remote Sensing* 65 (5): 457–67. https://doi.org/10.1016/j.isprsjprs.2010.06.001.

Beck, Scott M., Melissa R. McHale, and George R. Hess. 2016. "Beyond Impervious: Urban Land-Cover Pattern Variation and Implications for Watershed Management." *Environmental Management* 58 (1): 15–30. https://doi.org/10.1007/s00267-016-0700-8.

Bivand, Roger, Tim Keitt, and Barry Rowlingson. 2022. *Rgdal: Bindings for the "Geospatial" Data Abstraction Library*. https://CRAN.R-project.org/package=rgdal.

Brack, C. L. 2002. "Pollution Mitigation and Carbon Sequestration by an Urban Forest." *Environmental Pollution* 116 (March): S195–200. https://doi.org/10.1016/S0269-7491(01)00251-2.

Brewer, Will, Ashley Kale, Ashley Pruitt, Natlaia Rosario, Nelson Yaksic, Allison Wright, Mellone Long F., and Wayne Leftwich. 2015. "Pleasant Valley Visioning Plan." Clemson University. https://www.greenvillesc.gov/DocumentCenter/View/6742/Pleasant-Valley-Visioning-Plan-2016?bidId=.

Christensen, O. B., and J. H. Christensen. 2004. "Intensification of Extreme European Summer Precipitation in a Warmer Climate." *Global and Planetary Change*, Extreme climatic events, 44 (1): 107–17. https://doi.org/10.1016/j.gloplacha.2004.06.013.

"County of Greenville Zoning Ordinance." 2023. Greenville County. https://www.greenvillecounty.org/Zoning/pdf/OfficialZoningOrdinance.pdf.

Dielmo 3D. 2022. "Software User's Manual: AID-FOREST (Artificial Intelligence for Digital Forest) for the Automated Determination of Forest Stand Parameters." Elsevier. https://ars.els-cdn.com/content/image/1-s2.0-S1569843222002023-mmc1.docx.

Dong, Pinliang, and Qi Chen. 2017a. "Basics of LiDAR Data Processing." In *LiDAR Remote Sensing and Applications*. CRC Press.

———. 2017b. "Brief Overview of Remote Sensing." In *LiDAR Remote Sensing and Applications*. CRC Press.

Donovan, Geoffrey H., Jeffrey P. Prestemon, David T. Butry, Abigail R. Kaminski, and Vicente J. Monleon. 2021. "The Politics of Urban Trees: Tree Planting Is Associated with Gentrification in Portland, Oregon." *Forest Policy and Economics* 124 (March): 102387. https://doi.org/10.1016/j.forpol.2020.102387.

Drillet, Zuzana, Tze Kwan Fung, Rachel Ai Ting Leong, Uma Sachidhanandam, Peter Edwards, and Daniel Richards. 2020. "Urban Vegetation Types Are Not Perceived Equally in Providing Ecosystem Services and Disservices." *Sustainability* 12 (5): 2076. https://doi.org/10.3390/su12052076.

Dwyer, Mark, and Robert Miller. 1999. "Using GIS to Assess Urban Tree Canopy Benefits and Surrounding Greenspace Distributions." *Arboriculture & Urban Forestry* 25 (2): 102–7. https://doi.org/10.48044/jauf.1999.016.

Edgar, Christopher B, David J Nowak, Mark A Majewsky, Tonya W Lister, James A Westfall, and Nancy F Sonti. 2021. "Strategic National Urban Forest Inventory for the United States." *Journal of Forestry* 119 (1): 86–95. https://doi.org/10.1093/jofore/fvaa047.

Elmqvist, Thomas, Franz Gatzweiler, Elisabet Lindgren, and Jieling Liu. 2019. "Resilience Management for Healthy Cities in a Changing Climate." In *Biodiversity and Health in the Face of Climate Change*, edited by Melissa R. Marselle, Jutta Stadler, Horst Korn, Katherine N. Irvine, and Aletta Bonn, 411–24. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-02318-8_18.

Esri. 2022a. *Arcgisbinding: Bindings for ArcGIS*. http://esri.com.

———. 2022b. "ArcPy Documentation." ArcGIS Developers. 2022. https://developers.arcgis.com/documentation/arcgis-add-ins-and-automation/arcpy/.

Fowler, Hayley J., Haider Ali, Richard P. Allan, Nikolina Ban, Renaud Barbero, Peter Berg, Stephen Blenkinsop, et al. 2021. "Towards Advancing Scientific Knowledge of Climate Change Impacts on Short-Duration Rainfall Extremes." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 379 (2195): 20190542. https://doi.org/10.1098/rsta.2019.0542.

"GIS Data." n.d. City of Greenville, SC. Accessed April 27, 2023. https://www.greenvillesc.gov/364/Access-GIS-Data.

"GIS Data Access." n.d. Greenville County GIS. Accessed October 21, 2022. https://www.gcgis.org/AccessDistribution.html.

Greenville, SC. 2021. "Plant GVL: City of Greenville Tree Campaign." January 25, 2021. https://www.greenvillesc.gov/AgendaCenter/ViewFile/Item/7939?fileID=37015.

Greenville, SC, PlanIT Geo, and TreePlotter. n.d. “TreePlotter Inventory for Greenville, SC.” Accessed October 21, 2022. https://pg-cloud.com/GreenvilleSC/.

“Greenville’s Trees.” n.d. Greenville, SC. Accessed October 21, 2022. https://www.greenvillesc.gov/744/Greenvilles-Trees.

Hilbert, Deborah R., Andrew K. Koeser, Lara A. Roman, Keir Hamilton, Shawn M. Landry, Richard J. Hauer, Haley Campanella, Drew McLean, Michael Andreu, and Hector Perez. 2019. “Development Practices and Ordinances Predict Inter-City Variation in Florida Urban Tree Canopy Coverage.” *Landscape and Urban Planning* 190 (October): 103603. https://doi.org/10.1016/j.landurbplan.2019.103603.

Hill, Elizabeth, Jeffrey H. Dorfman, and Elizabeth Kramer. 2010. “Evaluating the Impact of Government Land Use Policies on Tree Canopy Coverage.” *Land Use Policy*, Forest transitions, 27 (2): 407–14. https://doi.org/10.1016/j.landusepol.2009.05.007.

Hobbie, Sarah E., Jacques C. Finlay, Benjamin D. Janke, Daniel A. Nidzgorski, Dylan B. Millet, and Lawrence A. Baker. 2017. “Contrasting Nitrogen and Phosphorus Budgets in Urban Watersheds and Implications for Managing Urban Water Pollution.” *Proceedings of the National Academy of Sciences* 114 (16): 4177–82. https://doi.org/10.1073/pnas.1618536114.

Holgado-Barco, Alberto, Diego Gonzalez-Aguilera, Pedro Arias-Sanchez, and Joaquín Martinez-Sanchez. 2014. “An Automated Approach to Vertical Road Characterisation Using Mobile LiDAR Systems: Longitudinal Profiles and Cross-Sections.” *ISPRS Journal of Photogrammetry and Remote Sensing* 96 (October): 28–37. https://doi.org/10.1016/j.isprsjprs.2014.06.017.

Hsu, Angel, Glenn Sheriff, Tirthankar Chakraborty, and Diego Manya. 2021. “Disproportionate Exposure to Urban Heat Island Intensity across Major US Cities.” *Nature Communications* 12 (1): 2721. https://doi.org/10.1038/s41467-021-22799-5.

Hughes, Conor. 2023. “Greenville’s Black Communities Disappearing as New Money Flows In.” Post and Courier. January 12, 2023. https://www.postandcourier.com/greenville/news/greenvilles-black-communities-disappearing-as-new-money-flows-in/article_fd1172d6-910e-11ed-8b52-07e530099054.html.

Hunter, Ruth F., Anne Cleary, and Matthias Braubach. 2019. “Environmental, Health and Equity Effects of Urban Green Space Interventions.” In *Biodiversity and Health in the Face of Climate Change*, edited by Melissa R. Marselle, Jutta Stadler, Horst Korn, Katherine N. Irvine, and Aletta Bonn, 381–409. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-02318-8_17.

Janke, Benjamin D., Jacques C. Finlay, and Sarah E. Hobbie. 2017. "Trees and Streets as Drivers of Urban Stormwater Nutrient Pollution." *Environmental Science & Technology* 51 (17): 9569–79. https://doi.org/10.1021/acs.est.7b02225.

Keeler, Bonnie L., Perrine Hamel, Timon McPhearson, Maike H. Hamann, Marie L. Donahue, Kelly A. Meza Prado, Katie K. Arkema, et al. 2019. "Social-Ecological and Technological Factors Moderate the Value of Urban Nature." *Nature Sustainability* 2 (1): 29–38. https://doi.org/10.1038/s41893-018-0202-1.

Klobucar, Blaz, Neil Sang, and Thomas B. Randrup. 2021. "Comparing Ground and Remotely Sensed Measurements of Urban Tree Canopy in Private Residential Property." *Trees, Forests and People* 5 (September): 100114. https://doi.org/10.1016/j.tfp.2021.100114.

Kolb, Ken, Mike Winiski, Sam Hayes, Catherine Lippert, and Sydney Andersen. 2023. "Racial Displacement in Greenville, SC." Storymap. ArcGIS StoryMaps. January 8, 2023. https://storymaps.arcgis.com/stories/ec193fcd3f194bc0a7f8353f69f24aa8.

Kontokosta, Constantine E., Vincent J. Reina, and Bartosz Bonczak. 2020. "Energy Cost Burdens for Low-Income and Minority Households." *Journal of the American Planning Association* 86 (1): 89–105. https://doi.org/10.1080/01944363.2019.1647446.

Lambers, Karsten. 2018. "Airborne and Spaceborne Remote Sensing and Digital Image Analysis in Archaeology." In *Digital Geoarchaeology: New Techniques for Interdisciplinary Human-Environmental Research*, edited by Christoph Siart, Markus Forbriger, and Olaf Bubenzer, 109–22. Natural Science in Archaeology. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-25316-9_7.

Li, Liqing. 2019. "The Effect of Urban Tree Planting on Residential Property Values and Gentrification." University of Illinois at Urbana-Champaign. https://www.liqing-li.com/publication/NYCtree.pdf.

Livesley, S. J., E. G. McPherson, and C. Calfapietra. 2016. "The Urban Forest and Ecosystem Services: Impacts on Urban Water, Heat, and Pollution Cycles at the Tree, Street, and City Scale." *Journal of Environmental Quality* 45 (1): 119–24. https://doi.org/10.2134/jeq2015.11.0567.

López Serrano, F. R., E. Rubio, F. A. García Morote, M. Andrés Abellán, M. I. Picazo Córdoba, F. García Saucedo, E. Martínez García, et al. 2022. "Artificial Intelligence-Based Software (AID-FOREST) for Tree Detection: A New Framework for Fast and Accurate Forest Inventorying Using LiDAR Point Clouds." *International Journal of Applied Earth Observation and Geoinformation* 113 (September): 103014. https://doi.org/10.1016/j.jag.2022.103014.

Luber, George, and Michael McGeehin. 2008. “Climate Change and Extreme Heat Events.” *American Journal of Preventive Medicine*, Theme Issue: Climate Change and the Health of the Public, 35 (5): 429–35. https://doi.org/10.1016/j.amepre.2008.08.021.

Manson, Steven, Schroeder, Jonathan, Van Riper, David, Kugler, Tracy, and Ruggles, Steven. 2022. “National Historical Geographic Information System: Version 17.0.” Minneapolis, MN: IPUMS. https://doi.org/10.18128/D050.V17.0.

Münzinger, Markus, Nikolas Prechtel, and Martin Behnisch. 2022. “Mapping the Urban Forest in Detail: From LiDAR Point Clouds to 3D Tree Models.” *Urban Forestry & Urban Greening* 74 (August): 127637. https://doi.org/10.1016/j.ufug.2022.127637.

NAIP. 2021. “Greenville County, South Carolina Color Infrared Imagery.” National Agricultural Imagery Program. Ortho_1-1_hc_s_sc045_2021_1. https://nrcs.app.box.com/v/naip/file/897162887486.

National Geospatial Program, and United States Geological Survey. 2018. “Lidar Base Specification.” 11-B4. Techniques and Methods. United States Deportment of the Interior. https://pubs.usgs.gov/tm/11b4/pdf/tm11-B4.pdf.

“Neighborhood Plans.” n.d. Greenville, SC. Accessed April 17, 2023. https://www.greenvillesc.gov/250/Neighborhood-Plans.

“Ordinance No. 2021-07.” 2021. Greenvile, SC. https://www.greenvillesc.gov/DocumentCenter/View/16229/City-of-Greenville-Tree-Protection-Ordinance?bidId=.

Pebesma, Edzer. 2018. “Simple Features for R: Standardized Support for Spatial Vector Data.” *The R Journal* 10 (1): 439–46. https://doi.org/10.32614/RJ-2018-009.

Pike, Kaitlyn, Keith O’Herrin, Christie Klimas, and Jess Vogt. 2021. “Tree Preservation during Construction: An Evaluation of a Comprehensive Municipal Tree Ordinance.” *Urban Forestry & Urban Greening* 57 (January): 126914. https://doi.org/10.1016/j.ufug.2020.126914.

Plowright, Andrew. (2016) 2022. “ForestTools: Tools for Analyzing Remotely Sensed Forest Data.” R. https://github.com/andrew-plowright/ForestTools.

Popescu, Sorin C, and Randolph H Wynne. 2004. “Seeing the Trees in the Forest: Using Lidar and Multispectral Data Fusion with Local Filtering and Variable Window Size for Estimating Tree Height.” *Photogrammetric Engineering & Remote Sensing* 5 (May): 589–604.

Prein, Andreas F., Roy M. Rasmussen, Kyoko Ikeda, Changhai Liu, Martyn P. Clark, and Greg J. Holland. 2017. “The Future Intensification of Hourly Precipitation Extremes.” *Nature Climate Change* 7 (1): 48–52. https://doi.org/10.1038/nclimate3168.

R Core Team. 2022. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Rigolon, Alessandro, and Jeremy Németh. 2020. "Green Gentrification or 'Just Green Enough': Do Park Location, Size and Function Affect Whether a Place Gentrifies or Not?" *Urban Studies* 57 (2): 402–20. https://doi.org/10.1177/0042098019849380.

Rodríguez-Puerta, Francisco, Carlos Barrera, Borja García, Fernando Pérez-Rodríguez, and Angel M. García-Pedrero. 2022. "Mapping Tree Canopy in Urban Environments Using Point Clouds from Airborne Laser Scanning and Street Level Imagery." *Sensors* 22 (9): 3269. https://doi.org/10.3390/s22093269.

Rötzer, T., A. Moser-Reischl, M. A. Rahman, R. Grote, S. Pauleit, and H. Pretzsch. 2021. "Modelling Urban Tree Growth and Ecosystem Services: Review and Perspectives." In *Progress in Botany Vol. 82*, edited by Francisco M. Cánovas, Ulrich Lüttge, María-Carmen Risueño, and Hans Pretzsch, 405–64. Progress in Botany. Cham: Springer International Publishing. https://doi.org/10.1007/124_2020_46.

Roussel, Jean-Romain, and David Auty. 2022. *Airborne LiDAR Data Manipulation and Visualization for Forestry Applications*. https://cran.r-project.org/package=lidR.

Roussel, Jean-Romain, David Auty, Nicholas C. Coops, Piotr Tompalski, Tristan R. H. Goodbody, Andrew Sánchez Meador, Jean-François Bourdon, Florian de Boissieu, and Alexis Achim. 2020. "LidR: An R Package for Analysis of Airborne Laser Scanning (ALS) Data." *Remote Sensing of Environment* 251 (December): 112061. https://doi.org/10.1016/j.rse.2020.112061.

Rutherford, Austin, Brennan Hansley, Brittni Olesen, John Taylor, Thomas Trawick, and Zulfiya Abdukhalikova. 2016. "Greater Sullivan Neighborhood Visioning Plan Inventory." Clemson University. https://www.greenvillesc.gov/DocumentCenter/View/6740/Greater-Sullivan-Vision-Plan-2016?bidId=.

Sanusi, Ruzana, Denise Johnstone, Peter May, and Stephen J. Livesley. 2017. "Microclimate Benefits That Different Street Tree Species Provide to Sidewalk Pedestrians Relate to Differences in Plant Area Index." *Landscape and Urban Planning* 157 (January): 502–11. https://doi.org/10.1016/j.landurbplan.2016.08.010.

Schottland, Taj. 2019. "Parks as a Solution to Climate Change." *Parks and Recreation*, April 5, 2019.

Stone, Brian, Jeremy J. Hess, and Howard Frumkin. 2010. "Urban Form and Extreme Heat Events: Are Sprawling Cities More Vulnerable to Climate Change Than Compact Cities?" *Environmental Health Perspectives* 118 (10): 1425–28. https://doi.org/10.1289/ehp.0901879.

Tanhuanpää, Topi, Xiaowei Yu, Ville Luoma, Ninni Saarinen, Juha Raisio, Juha Hyyppä, Timo Kumpula, and Markus Holopainen. 2019. “Effect of Canopy Structure on the Performance of Tree Mapping Methods in Urban Parks.” *Urban Forestry & Urban Greening* 44 (August): 126441. https://doi.org/10.1016/j.ufug.2019.126441.

Tarpley, J. D., S. R. Schneider, and R. L. Money. 1984. “Global Vegetation Indices from the NOAA-7 Meteorological Satellite.” *Journal of Applied Meteorology and Climatology* 23 (3): 491–94. https://doi.org/10.1175/1520-0450(1984)023<0491:GVIFTN>2.0.CO;2.

“Tree Canopy in Greenville, SC.” 2023. OurTrees 1.4.5. April 27, 2023. https://ourtrees.itreetools.org/#/report?longitude=-82.3940104&latitude=34.85261759999999.

U.S. Forest Service, American Forests, and National Association of Regional Councils. n.d. “Vibrant Cities Lab : Resources for Urban Forestry, Trees, and Green Infrastructure.” Accessed February 12, 2023. https://www.vibrantcitieslab.com.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

Wickham, Hadley, Romain François, Lionel Henry, and Kirill Müller. 2022. *Dplyr: A Grammar of Data Manipulation*. https://CRAN.R-project.org/package=dplyr.

Yang, Ping, Daniel P. Ames, André Fonseca, Danny Anderson, Rupesh Shrestha, Nancy F. Glenn, and Yang Cao. 2014. “What Is the Effect of LiDAR-Derived DEM Resolution on Large-Scale Watershed Model Results?” *Environmental Modelling & Software* 58 (August): 48–57. https://doi.org/10.1016/j.envsoft.2014.04.005.

Yun, Ting, Kang Jiang, Guangchao Li, Markus P. Eichhorn, Jiangchuan Fan, Fangzhou Liu, Bangqian Chen, Feng An, and Lin Cao. 2021. “Individual Tree Crown Segmentation from Airborne LiDAR Data Using a Novel Gaussian Filter and Energy Function Minimization-Based Approach.” *Remote Sensing of Environment* 256 (April): 112307. https://doi.org/10.1016/j.rse.2021.112307.

Zhang, Yihuan, Jun Wang, Xiaonian Wang, and John M. Dolan. 2018. “Road-Segmentation-Based Curb Detection Method for Self-Driving via a 3D-LiDAR Sensor.” *IEEE Transactions on Intelligent Transportation Systems* 19 (12): 3981–91. https://doi.org/10.1109/TITS.2018.2789462.

Ziter, Carly D., Eric J. Pedersen, Christopher J. Kucharik, and Monica G. Turner. 2019. “Scale-Dependent Interactions between Tree Canopy Cover and Impervious Surfaces Reduce Daytime Urban Heat during Summer.” *Proceedings of the National Academy of Sciences* 116 (15): 7575–80. https://doi.org/10.1073/pnas.1817561116.